What Has Been Proved So Far on the Erdős–Straus Conjecture Under a Human-Directed, Two-System Research Protocol

Narrow theorems, machine-certified statements, and the exact residual gap

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Note on authorship and AI assistance.
The research reported here was directed by the author. The mathematics, computations and formal proofs were produced by two large language model systems, Claude (Anthropic) and GPT (OpenAI), working in separate conversations under the author's written protocol; the author made every launch, labelling and publication decision, performed the independent reruns, and held sole write access to the verification repositories. Version 1 of this draft was written by Claude from the project's governing records on 29 October 2026; a parallel draft was commissioned from GPT and received the same evening. Claude reviewed GPT's draft (its seven new energy-route theorems were checked by an independently written program, Appendix B), compared the two, and wrote version 2; the energy-route theorems of §4.7 are GPT's, credited as such. Version 3 (30 October) adds the refutation of Conjecture Q, found by Claude in response to the author's instruction to attack it computationally, confirmed by GPT with its own sealed implementation, and replicated by the author; and it records the public deposit of every jointly validated artefact. Every claim below carries the evidence label it held in the project charter on the date above. Nothing in this paper is a proof of the Erdős–Straus conjecture. Version 6 (30 October) adopts the corrections of GPT's review of version 5 — the scope of the two-free-prime impossibility statement, the status of the higher-factor barrier, the accounting of Table 3, the definition and counting of the extended energy test, and the interpretation of the three methods' failures — each adopted after Claude's independent check; reconciles Table 3 from the row-level Round 1–3 archives, recovered by GPT and audited by Claude; and adds §4.10, an exact asymptotic for a published divisor weight, with its verification status stated exactly. Version 6.2 (30 October) renumbers the figures in order of appearance, removes the title printed inside the p = 577 figure so that every figure is titled only by its caption, restores the Figure 1 ordering sentence, and adds Appendix C, a lexicon translating the paper's technical vocabulary into plain language. Version 6.3 (31 October) adds §4.11, the Protean Z round proposed by the author and executed by GPT on 30 October: an exact centered-witness lemma, a Kneser-type sufficient criterion, two "simple witness" hypotheses of which one (S1) is refuted by Claude at p = 1,977,361 and the other (S2) holds at every residual prime below 10^7, two further refuted proposals, a closed Galton-type analysis with an exact negative, an empirical per-shift success profile replicated by the author, and a calibration of the proposed failure-packet model. It also records two corrections to Claude's statements made by GPT, and the Charter's development-range rule adopted on 31 October. Version 6.5 (31 October) is the version of record deposited on SSRN; it adds seven entries to the lexicon of Appendix C (shift, channel, energy, finite energy, witness, sieve, weight), corrects the version label on the title page, and names the location of this PDF in the public repository. No mathematical statement or evidence label changed between versions 6.3 and 6.5.

(Public record: github.com/whouxe82/es-lean-certification)

Abstract

The Erdős–Straus conjecture asserts that $4/n = 1/x + 1/y + 1/z$ has a solution in positive integers for every $n \geq 2$. It is known for every residue class modulo 840 except the six square classes 1, 121, 169, 289, 361, 529, and a classical obstruction shows that no polynomial identity can settle those classes. This paper reports, with explicit evidence labels, what a human-directed research line run in parallel by two large language model systems proved about the conjecture between 27 and 29 October 2026, under a protocol requiring sealed statement-first exchange, adversarial reconstruction, independent validators, exact arithmetic, and formal certification. The results are of four kinds. (i) Exact, bounded-size residue criteria for the two divisor channels at a shift d: a one-free-prime criterion A5′ (success at d is decided by p mod d lying in a set S(m, d) of at most $2\tau(m^2)$ residues), a two-free-prime criterion, and a chain-saturation criterion; together these certify 49,305 of the 50,515 archived successes (97.60%) that an earlier spectral criterion had missed, leaving 1,210 rows in which at least three prime factors are quadratic non-residues modulo d; on the two-free-prime domain no criterion in p mod d alone can decide success, and whether a bounded criterion that retains the individual prime residues extends further is OPEN. (ii) Lean 4 declarations, rebuilt independently by GitHub Actions on the author's account, certifying an integral matching theorem (the graph of positive solution triples at every prime $p \equiv 1 \pmod 4$ under the bridge map has degree at most one and no self-loops), a complete descent classification for the fourth-iterate return (B4), exact integer-return gates, an abstract finite-energy forcing theorem, and an infinite progression of H_N witnesses, with Dirichlet's theorem and quadratic reciprocity imported as audited dependencies. (iii) An exact local theory of an "energy route": a closed-form integer minimum for the target-avoiding energy of a divisor histogram, a collision form of the forcing test, a complete classification of the inputs on which avoidance is impossible, uniform families covering the 18 non-square classes modulo 840, and a correctly scoped theorem that no fixed finite menu of shifts supplies an admissible input for every prime; the universal statement (Conjecture Q), which held in a two-implementation pilot to 1873, is then refuted: an exhaustive decision at every prime $p \equiv 1 \pmod{24}$ below 10^6 finds 25 counterexamples, the smallest $p = 66{,}361$, all in the residual family, confirmed by the two systems' independent implementations, by a Proposition-15 certificate, and by the author's replication; the extension without the specification's non-residue restriction fails at $p = 944{,}329$. The Erdős–Straus conjecture holds at every failing prime: the refutation concerns the route, which nevertheless certifies all primes outside the residual family and 98.3% of those within it below 10^6. (iv) A conditional density theorem (every shift $d \equiv 3 \pmod 4$ fails, separately, on $\gg N/(\log N)^{3/2}$ primes, conditional on a 2026 theorem on primes represented by shifted quadratic forms) and an exact reduction of the residual classes to a pointwise sieve covering statement, measured by a census showing that every one of the 444,820 primes $p \equiv 1 \pmod{24}$ below $6 \cdot 10^7$ is settled by a shift $d \leq 107$; the literature check performed for this version found that this statistic and its record primes were published by Ionascu and Wilson in 2011, so the census is a replication, not a finding. (v) An exact asymptotic for the divisor weight $W(J) = \Sigma\, 1/\varphi(4ua)$ over coprime pairs $u, a \leq J$ of Dahan's Lemma 4.22: $W(J) = (9/2\pi^2)(\log J)^2 + O(\log J)$, with $9/2\pi^2 \approx 0.456$ against the proved lower constant $3/2\pi^2$ and the constant of at least 0.45 that paper asks for; proved by GPT in two derivations, verified by Claude by a different route and numerically, originality under check. (vi) A round of "simple witness" hypotheses: every residual prime should have, within an explicit window of shifts of size about $(\log_2 p)^2$, a solution whose divisor witness changes the exponent of at most one (S1) or two (S2) prime factors of x. S1 is refuted at $p = 1{,}977{,}361$ by complete enumeration of the window; S2 holds at all 20,513 residual

primes below 10^7 with largest first shift 151, and an empirical per-shift profile (success probability about 12/j at shift index j) shows why S2 can be neither refuted by search nor proved by the sieve. A Galton-type analysis of divisor distributions is closed with an exact negative: two residual primes sharing x = 494,392 have identical divisor, support and size distributions and 0 versus 5 witnesses. The paper argues that the mathematical gain is modest and precisely bounded — new exact criteria of possibly original form, certified structural theorems whose hypotheses are explicit, and a sharpened statement of where the conjecture is stuck — while the methodological gain is the labelled, reproducible record itself. Every jointly validated artifact — programs, records, certificates and replication receipts — is deposited in a public repository.

Keywords: Erdős–Straus conjecture; Egyptian fractions; unit fractions; quadratic residues; Jacobi symbol; sieve methods; Lean 4; formal verification; human–AI collaboration; large language models; research governance.

1. Introduction

Paul Erdős and Ernst Straus conjectured in 1948 that for every integer $n \geq 2$ the fraction $4/n$ is a sum of three unit fractions. It suffices to prove the statement for primes, and elementary identities dispose of every prime outside a finite list of residue classes: Mordell showed that the conjecture holds for all n except possibly those congruent to 1, 121, 169, 289, 361 or 529 modulo 840, which are exactly the squares of units modulo 840 [Mordell 1969]. A theorem reported by Mordell and by Schinzel shows that no identity of the form $4/n = 1/f(n) + 1/g(n) + 1/h(n)$ with polynomial f, g, h can be valid on a whole class $n \equiv r \pmod m$ when r is a square modulo m [Elsholtz–Tao 2013, §1]. The obstruction concerns a fixed triple of rational-coefficient polynomials valid on a whole class; it says nothing about rules that choose denominators adaptively or by factorization. The six residual classes therefore resist every method of that uniform polynomial form by necessity, not by accident, and the conjecture has been verified computationally far beyond 10^{17} while remaining open.

This paper is a progress report of an unusual kind. In October 2026 the author, a humanities Ph.D. with no mathematical training beyond the undergraduate level, directed two large language model systems — Claude and GPT — in a structured research line on the conjecture, under a written protocol (the ES XYZ Research Charter) whose purpose was to obtain, in each round, *a completely proved, precisely scoped contribution beyond the established baseline: a limited theorem, a useful negative result, or an honest inconclusive report*. A proof of the full conjecture was named in the charter as an ambition, never as a deliverable. The protocol required every claim to carry four fields: a mathematical status (PROVED, CONDITIONAL with the condition named, FINITE_VERIFIED with its bound, CONJECTURED, REFUTED with the witness kept, or INCONCLUSIVE), a review status, a novelty assessment, and a concrete statement of added scope. Formal verification in Lean 4 had its own three-valued label, and only a rebuild on the author's own GitHub account could move a statement to VERIFIED.

The paper has two audiences and two purposes. For mathematicians, it collects the theorems that survived adversarial review, with their exact hypotheses, their dependencies and the places where they stop, so that each can be checked, used or dismissed on its merits. For readers interested in how such systems can be made to do mathematics accountably, it documents what the protocol produced, what it caught (several false side-claims, one false finiteness step, one tautology presented as an observation), and what the record looks like when a participating system is withdrawn mid-report. Section 2 fixes notation and the classical criterion everything rests on. Section 3 describes the methodology. Section 4 states the findings, in descending order of evidential strength. Section 5 assesses their significance and Section 6 concludes. Proof sketches and reproduction details are in the appendices. Version 2 added the review of GPT's parallel draft (§4.7) and a literature check that reclassified the channel-depth census of §4.6 as a replication of published work. Version 3 adds §4.8, the refutation of the energy route's universal conjecture, and §4.9, the record of its confirmation and public deposit. Version 6 corrects five statements of version 5 that outran their evidence (§4.2, §4.3, §4.8, §5.1, §5.4), reconciles the certification accounting of Table 3 from the recovered row archives, and adds §4.10. Version 6.3 adds §4.11 and the corresponding rows of Tables 4 and 7, a fifth row in Figure 5, and the development-range rule. Version 6.5 completes Appendix C and is the version deposited on SSRN and in the repository.

2. Background and notation

Throughout, p is a prime with $p \equiv 1 \pmod 4$. An *admissible shift* is an integer $d \equiv 3 \pmod 4$ with $0 < d < 2p$; composite d is allowed unless stated. For each admissible d put $x_d = (p + d)/4$, an integer in $(p/4, 3p/4)$ with $\gcd(x_d, d) = 1$. The notation is collected in Table 1.

Table 1. Notation and the evidence vocabulary used throughout.

Symbol	Meaning
p	prime, $p \equiv 1 \pmod 4$; "six-class" means $p \equiv 1, 121, 169, 289, 361$ or $529 \pmod{840}$, equivalently $p \equiv 1 \pmod{24}$ and p a quadratic residue modulo 5 and 7
d	admissible shift: $d \equiv 3 \pmod 4$, $0 < d < 2p$ (prime or composite)
x_d	$(p + d)/4$; the smallest p-free denominator of any solution lies in $(p/4, 3p/4)$ and determines $d = 4x - p$
$(\cdot/d)$	Jacobi symbol; for prime d the Legendre symbol
Channel I / II	existence of a divisor $a \mid x_d^2$ with $4a \equiv -1 \pmod d$ (Channel I, Type I solution: exactly one denominator divisible by p) or $a \equiv -x_d \pmod d$ (Channel II, Type II: two denominators divisible by p)
success / failure at d	some channel fires at d / neither does
QR-smooth	every prime factor r of x_d has $(r/d) = +1$
$\tau(n)$, $\omega(n)$	number of divisors; number of distinct prime factors
W4 archive	the project's development data: all 20,513 six-class primes $p < 10^7$ against the 15 prime shifts $d \le 107$; 138,221 known successes; never used as a holdout
B6 residual	the 50,515 archived successes not certified by an earlier spectral criterion T'_6; the population on which the residue criteria of §4.2 were measured
Labels	PROVED, CONDITIONAL (condition named), FINITE_VERIFIED (bound), CONJECTURED, REFUTED, INCONCLUSIVE; Lean: STATED, COMPILED, VERIFIED; novelty: KNOWN, EQUIVALENT_TO_KNOWN, POSSIBLE_NEW, NOT_ASSESSED

The whole line rests on one classical fact, re-proved within the project (Appendix A, Lemmas 1–2) and partly formalized.

> **Proposition 2.1 (two-channel divisor criterion; classical).** *Let $p \equiv 1 \pmod 4$ be prime and x an integer with $p/4 < x < 3p/4$, $d = 4x - p$. Then positive integers y, z with $4/p = 1/x + 1/y + 1/z$ exist if and only if some divisor a of x^2 satisfies $a \equiv -x \pmod d$ (Channel II; both y and z divisible by p) or $4a \equiv -1 \pmod d$ (Channel I; exactly one of y, z divisible by p). The return maps are explicit (Appendix A.1): in Channel II, $y = p(a + x)/d$ and $z = p(x^2/a + x)/d$; in Channel I, with $a' = x^2/a$, $y = (a' + px)/d$ and $z = p(px^2/a' + x)/d$.*

The Channel I condition is stated in two equivalent forms in the project records, $a \equiv -px \pmod d$ and $4a \equiv -1 \pmod d$; they correspond under the involution $a \mapsto x^2/a$ of the divisors of x^2, because $p \equiv 4x \pmod d$ gives $-px \equiv -4x^2$, and $x^2/a \equiv -4^{-1}$ exactly when $a \equiv -4x^2$.

Two consequences organise everything below. First, whether a channel fires at d depends only on the multiset of residues modulo d of the divisors of x_d^2, hence only on the prime

factorization of a shifted prime (p + d)/4 — a quantity no polynomial in p controls. Second, since every solution has a smallest p-free denominator in (p/4, 3p/4), the conjecture at p is equivalent to the statement that some admissible d fires; the least such d is the *channel depth* D(p) of §4.6.

3. Methodology

3.1 The XYZ selection rule

Every proposed task was written as a card with three parts: X, an exact established statement and its domain; Y, the exact obstruction to extending it; Z, the change of question, representation, constraint or scale that addresses Y. Each card also stated a falsifiable consequence, an exact return map to positive integer denominators, the nearest known result, and a stop or revision condition. A larger numerical search alone was never an acceptable contribution, and a Z counted as a structural advance only if it introduced an explicit new invariant, compression or witness construction; a relabeled exhaustive divisor test was not a Z. The author's working hypothesis, stated in the charter and tested by the line, is that frontier language models perform on the X and Y axes and that the Z component — the frame change a breakthrough requires — is what has to be demanded of them explicitly and may not be forthcoming.[1]

3.2 Two controllers, ten roles, sealed exchange

Two controller conversations, one per system, each coordinated ten role assignments: R01–R05 originating proposals and proofs, R06 exact computation with preregistered populations and predicates, R07 literature and originality from primary sources, R08 formal statement and Lean verification (no success claimed without a recorded build), R09 adversarial review that reconstructs an argument from its statement before reading the proof, and R10 an independent computational validator written before the producer's code is inspected. Role assignments were explicitly not a claim that ten independent instances ran; each run record states which agents were actually launched. Lane ownership was separated: GPT owned composite incompatibility

[1] The author has argued the structural form of this hypothesis in two earlier essays, in terms other than X/Y/Z. In "Here Lies the Standalone Frontier LLM: The Box Became Larger. But the Box Remained the Box" (SSRN 7118623, written 14 July 2026) the thesis is that no software system executing on a Turing-equivalent formal system — the base-2 digital computer being the present instance — can transcend the formal computational constraints of its human-engineered architecture; that this is a checkable claim, since mathematical logic shows how a proposition can be proved underivable from a given axiom set; and that until such evidence is produced, more capable post-LLM architectures should be read as enlarging the scope of engineered computation, not as transcending its foundational limits. In "Waiting for Gödel: The Inevitable Structural Break in the Scaling Program Positioned to Achieve Artificial General Intelligence" (SSRN 6740920, written 9 May 2026) the same limit is examined empirically: a one-page diagnostic table put to five leading systems exposed register competence under prompting in place of the structural reading the regime is bounded against, and a "Pre-Gödel Proof Exercise" states the distinction symbolically. Read against the present paper, these essays name the impasse the ES XYZ line met: the systems supplied exact X and Y work, caught each other's errors, and refuted their own conjecture, but the frame change the residual classes require — a pointwise sieve statement, or a certificate for firing channels on sparse-divisor primes — was not produced by any engineering of the systems or the protocol, and on the author's thesis could not be, because it is not located in the machine. A third essay, "Assessing the Design and Operational Integrity of Hyperscale Datacenters" (SSRN 7088419, written 9 July 2026), draws the related distinction between operational integrity (the machine computes what it was built to compute) and epistemic integrity (the output is true and warranted), locating the latter in a relation of answerability that the machine can occupy on one side only; the labelling and replication discipline of §3 is the practical form of that distinction.

obstructions, discrete compression, Lean execution and the shared Lean project; Claude owned positive compatibility, target-avoidance criteria, analytic shift obstructions and the review of GPT's build evidence. Exchange between systems passed through the author in two stages: statement-only archives first (so the receiving system's reviewers recorded an independent derivation or an explicit failed attempt before seeing any proof), then evidence archives with proofs, code and raw results. Files were sealed with SHA-256 hashes and timestamps; hashes establish byte identity, not truth or priority, and the charter says so.

3.3 Evidence rules and dependency rulings

Final decisions used exact integer or rational arithmetic; floating-point exploration was labelled; a positive certificate lists p, d, the distinguished x, channel, a, u, v, y, z and the identity check, and a negative search certificate states the whole domain checked with a completeness argument. Every count names its unit — (p, d) pairs, (p, d, channel) incidences, or distinct primes — and percentages of already-known successes are stated to measure certification by exact criteria, never progress toward the conjecture. The author ruled on dependencies: a result resting on a published theorem whose proof was not re-verified within the round is labelled CONDITIONAL on that theorem, never "proved relative to"; Dirichlet's theorem on primes in progressions (27 October) and quadratic reciprocity as formalized in Mathlib (28 October) were ruled *audited dependencies*, so results resting on them are labelled "depends on audited dependency" rather than CONDITIONAL. Open conjectures (Dickson, Schinzel H) were treated as a separate kind of uncertainty and never as dependencies. Withdrawn preprints were bibliographic leads only.

3.4 Formal verification and the author's independent rerun

Lean status took exactly three values: STATED (source written, no build), COMPILED (built and axiom-audited on one system's machine, with log, lake-manifest pins and verdicts preserved) and VERIFIED (the same pinned version rebuilt by GitHub Actions on the author's account with identical manifest and verdicts). On 28 October GPT delivered a pinned Lean 4.19.0/Mathlib package for the surviving structural theorems; the author created the public repository whouxe82/es-lean-certification, uploaded the package and a workflow file supplied by Claude, and GitHub Actions run #1 rebuilt it with a green result. The audit module walks every constant in the ES namespace, rejects any non-standard axiom and checks each theorem's axiom closure against a whitelist (propext, Classical.choice, Quot.sound); a separate source scan rejects sorry, admit, axiom, native_decide and kernel-bypass flags. Finite checks use kernel decide. Claude then audited, declaration by declaration, whether the package's prose claimed exactly what the declarations prove (§4.1). On 29 October the author found that GPT's GitHub connector had pushed a commit to the repository; he removed that access, cancelled all GPT subscriptions the same day after GPT's final session ended mid-report, and recorded the roster change in the charter. From that point the line continued with Claude as controller and implementer, the author as sole validator and committer, and a literature-search tool for citation checks.

Confirming a computational refutation (added 30 October). When one system's program finds a counterexample to a frozen conjecture, the protocol used for confirmation was: the other system reruns its own sealed implementation on the stated input, leaving the sealed files untouched (hash-verified before and after) and routing any needed change of bounds through a separately identified adapter; a second program that imports no sealed code and calls no optimizer validates the resulting certificate object by object; and the author reruns the finding

program himself. Three artefacts result — the finder's record, the confirmer's certificate, the director's output — and the label moves only when all three agree. The word "independent" is used throughout with its dimension named: a second run on another machine establishes reproducibility of one implementation; a separately written program tests dependence on an algorithm; a certificate checker that imports no search code removes trust in the search; a Lean build addresses a formal statement; and the claim-by-declaration audit addresses whether that statement is the intended one. These reinforce one another and do not substitute for one another.

3.5 Computation and budgets

Each exact experiment was budgeted (first run ≤ 5 CPU-minutes, ≤ 25 more for a surviving target, 90 CPU-minutes per system per round, memory ≤ min(5 GiB, half available)); actual CPU and wall time were recorded and model-inference time was never estimated. Validation populations were frozen only after a claim and method were frozen. The census of §4.6 and the witness checks of §4.5 were recomputed for this paper from a single standard-library Python program (Appendix B), in exact arithmetic, and agree with the project records to the last digit.

4. Findings

Figure 5 (end of this section) maps every result below to its evidence label. Results are grouped by strength: machine-certified statements first, then theorems proved and reviewed within the line, then certification accounting, conditional results, negative results, the non-polynomial reduction, the energy route's local theory, the refutation of its universal conjecture, the confirmation and deposit record, and finally (§4.10, added in version 6) an exact asymptotic for a published divisor weight, and (§4.11, added in version 6.3) the Protean Z round of simple-witness hypotheses and the failure-packet model, both shown at the base of the map.

4.1 Statements certified in Lean 4 and independently rebuilt

The following statements are Lean-certified at commit a92f656 of whouxe82/es-lean-certification, rebuilt in GitHub Actions run #1 on 28 October 2026. The package contains 191 authored theorem and lemma commands in 12 modules; every one depends only on the three standard axioms. The three open propositions — the conjecture itself (ES.ErdosStraus) and two sufficient hypotheses H_D and H_N — are definitions, never asserted. The claim-by-declaration review found 20 of 24 prose claims fully supported, 3 supported with a stated qualification, 1 explicitly not formalized, and none overstated. The setting is a normalized Type II source (δ, r, s, t) at $p = 4k + 1$ with $x = \delta rs$, $y = p\delta rt$, $z = prst$, shift d with $dt = \delta + s$ and $p + d = 4\delta rs$, and C defined by $4C = pd + 1$; T is the rational "bridge" map that carries one solution triple to another, with B1 and B4 the first and fourth iterate returns.

Table 2. Lean-certified statements and their scope qualifications.

Certified statement	Declarations (module)	Hypotheses and scope	Qualification
Reduction to primes $p \equiv 1 \pmod 4$; $p \equiv 3 \pmod 4$ and $p = 2$ handled explicitly	erdosStraus_iff_primes, one_mod_four_suffices (Basic)	none	—
Prime-input solution is Type I or Type II; not all three denominators divisible by p	typeI_or_typeII, not_all_divisible (Matching)	p prime, $p > 4$	—

Certified statement	Declarations (module)	Hypotheses and scope	Qualification
Type II normalization is derived, not assumed: any solution with p | y, p | z, p ∤ x has x = δrs, y = pδrt, z = prst	normalize_typeII (Normalization)	p prime	—
Integral matching theorem: on positive solution triples up to permutation, with edges the integer images of T in any orientation, every vertex has at most one neighbour and no vertex is adjacent to itself	solutionGraph_neighbors_unique, degree_at_most_one, no_self_edge (MatchingGraph)	p prime, p = 4k + 1; no existence premise	vertices are multisets, so multiplicity is retained; the only result with no source hypothesis
Type I source has no integral outgoing edge	typeI_no_outgoing (Matching)	p prime, p = 4k + 1	no reciprocity in the proof chain
B1 integer return ⇔ δt | k; B4 return (tC/s, tC/δ, pC) is the literal fourth iterate and is a positive integer solution ⇔ 4δs | $d^2 - 1$	b1_integer_return_iff, b4_four_iterates, b4_integer_return_iff (Bridges)	normalized Type II source; positivity carried inside the definitions	says nothing about integrality of the three intermediate points
B4 descent classification: every B4-eligible normalized source has t = 1 and δ odd; if δ is a power of two then δ = t = 1 and the B4 endpoint equals the B1 endpoint	b4_source_restriction, b4_dyadic_no_rescue, b4_dyadic_endpoints_equal (Descent)	no primality premise	—
Every positive ordered terminal pair (a, b) with $(a + b)^2 = 4mab + 1$ is generated from (1, c) by (a, b) ↦ (b, cb − a); the mod-4 sum selects even indices	norm_recurrence_complete, shapePair_allowed_iff_even (Descent)	0 < m, 0 < a ≤ b	—
H_D ⇒ ES and H_N ⇒ ES	hd_implies_es, hn_implies_es (Basic)	H_D, H_N as explicit hypotheses (definitions over primes p ≡ 1 mod 4)	conditional implications; premises not proved
Finite forcing theorem: a compatible histogram with energy strictly below the finite avoidance minimum occupies a target bin	energy_lt_avoidanceMinimum_forces (Forcing)	abstract: index set, central bin and targets are parameters	arithmetic instantiation (divisor involution, parity of fixed bins) is a separate, unformalized obligation
H/H^2 is the finest group quotient preserved by complementation	preservesCosets_iff_squares_le (Forcing)	commutative group, subgroup	—
p = 577: the true divisor histogram of 154^2 has energy 12, avoidance minimum 14, so a divisor ≡ 2 (mod 39) is forced	counts577_energy, avoidanceMinimum577_eq, divisor577_target_exists (Forcing)	concrete numbers only	mechanism certified; the conclusion is satisfied trivially by a = 2; the cell is relative to squares of the full unit group mod 39, not the subgroup ⟨2, 7, 11⟩ (they coincide here, unproved in general)
Family p = 528n + 505: positive source and endpoint for every n; endpoint a non-residue and an H_N	endpoint_nonresidue, endpoint_HN_witness, nonresidue_family_infinite,	depends on audited dependencies: Mathlib Dirichlet and	a progression theorem — a regression check, not progress on

Certified statement	Declarations (module)	Hypotheses and scope	Qualification
witness at every prime member; infinitely many prime members, in each of the six classes mod 840	six_classes_infinite (Families)	reciprocity	universality

Table 2 lists every statement that may carry the certification label, with the declarations that prove it, the hypotheses under which it holds, and the qualifications attached by the claim-by-declaration review of 29 October 2026. The qualification column is the important one: it records where the prose of the certification package was found to say slightly more than the Lean, and those points were corrected before this paper was written.

Not certified, and not to be described as such: the general B1 character-transfer argument; the reduction of H_D/H_N from $p \equiv 1 \pmod 4$ to $p \equiv 1 \pmod{24}$; the target-to-ES reconstruction in general; the parity of fixed bins for a general divisor histogram; and Q(p) for any p (§4.7). Separately, GPT's earlier local build of 27 October (Lean 4.35.0) compiled 44 declarations including the forward two-channel soundness and Claude's Lemma U1; that build holds the label COMPILED, since it was never rebuilt on the author's account.

4.2 Theorems proved and reviewed within the line: exact residue criteria

The theorems of this subsection are PROVED by elementary arguments, were reconstructed blind by the adversarial reviewer from their statements, were verified with zero mismatches on development and validation data by independently written validators, and were accepted by the counterpart system's review. None is formally verified. Their novelty is labelled POSSIBLE_NEW: the mechanism (Jacobi-parity forcing on divisors) is classical, the statements were not found in the sources consulted (Mordell; Elsholtz–Tao; López; Bello-Hernández et al.; Dahan; Jiang; Xu; Dyachenko; Cao–Xu), and absence from all literature cannot be certified by the method used. The engine is the parity observation that a divisor of x_d^2 has Jacobi symbol -1 modulo d if and only if it contains an odd total exponent of primes with symbol -1, together with the fact that, since $d \equiv 3 \pmod 4$, the Channel I target -4^{-1} has Jacobi symbol -1 while the Channel II target $-x_d$ has symbol $-(x_d/d)$.

Lemma U1 (QR-smooth implies failure; unconditional). *For every admissible d, prime or composite, and every prime $p \equiv 1 \pmod 4$: if every prime factor r of x_d has $(r/d) = +1$, then p fails at d. [PROVED; Lean COMPILED in GPT run_002; novelty EQUIVALENT_TO_KNOWN (Mordell-type). The composite-d form matters: on the validation range 21,989 rows were Jacobi-smooth but not Legendre-smooth at the prime factors of d, and all failed.]*

Lemma R (reciprocity transfer). *For every prime $p \equiv 1 \pmod 4$, admissible d and prime $r \mid x_d$ (r = 2 included): $(r/d) = (r/p)$, Jacobi on the left and Legendre on the right; hence $(x_d/d) = (d/p) = (p/d)$. No hypothesis on p mod 8. [PROVED; KNOWN.]*

Theorem A5′ (one-free-prime criterion). *Let d be admissible and suppose $x_d = m \cdot s$ where s is a prime not dividing m, $(s/d) = -1$, and every prime factor of m has Jacobi symbol +1 modulo d. Put $R = m^{-1} \cdot \{a \bmod d : a \mid m^2\} \subset (\mathbb{Z}/d\mathbb{Z})^\times$. Then Channel I fires at d if and only if $p \bmod d \in -R$, and Channel II fires if and only if $p \bmod d \in -4mR$. In particular success at d is decided by $p \bmod d$ lying in the set $S(m, d) = -R \cup -4mR$, of*

size at most $2\tau(m^2)$, independent of p. [PROVED for prime and composite d; accepted by GPT review; 0 mismatches on 237,232 development rows and on the frozen validation range; POSSIBLE_NEW in this generality.]

The proof is short enough to give (Appendix A.2). The bound $2\tau(m^2)$ is immediate from the two-set description; the project's Round 1 statement carried the weaker $3\tau(m^2)$, and the reduction to $2\tau(m^2)$ was one of the counterpart's requested sharpenings. A corollary explains a phenomenon visible in the archive: for prime d, when the one-free-prime hypothesis holds, every success occurs at a "flip" shift — one with $(d/p) = -1$ — because $(x_d/d) = (d/p)$ by Lemma R and x_d contains exactly one non-residue prime.

Theorem C3-01 (two-free-prime criterion). *Let d be admissible and $x_d = m \cdot s_1 \cdot s_2$ with $s_1 \neq s_2$ primes, $(s_1/d) = (s_2/d) = -1$, $s_i \nmid m$, and every prime factor of m Jacobi +1 modulo d. With R as in A5′, success at d holds if and only if the pair of residues $\{\bar{s}_1, \bar{s}_2\}$ meets the explicit set $W = -R \cup -(4m)^{-1}R \cup -\bar{p}^{-1}R$, where $\bar{p} = p \bmod d$; $|W| \leq 3\tau(m^2)$. The first two parts are A5′ lifted to the pair (pure residue conditions); the third is a joint condition on the pair and fires in Channel I only. Moreover, on the domain of the sealed proof — prime d with squarefree two-free fibres — no criterion for this family can depend on p mod d alone: there every quadratic-residue class of $\bar{p}$ carries both outcomes at residue level. That impossibility statement does not extend to every admissible d. At $d = 3$ every input of the family succeeds: $x_d \equiv 1 \pmod 3$ because every prime of m is $\equiv 1$ and $s_1 s_2 \equiv 1 \pmod 3$, while each $s_i \equiv 2 \pmod 3$ is itself the residue $-x_d = -px_d = 2$ that both channels target; $p = 1009 \equiv 169 \pmod{840}$, $x_3 = 253 = 11 \cdot 23$, is an instance inside the six classes. The exact exceptional moduli of the impossibility clause are therefore those excluded in its sealed proof, and version 5's unrestricted wording is withdrawn. [PROVED; accepted by GPT review with 14 wording and unit corrections; 0 mismatches on all 62,925 two-free rows of the archive; certifies 14,761 of the 16,646 residual rows then uncertified, exactly the preregistered count.]*

Theorem CS (chain saturation). *Write $x_d = \prod r_i^{e_i}$, $G = (\mathbb{Z}/d\mathbb{Z})^\times$, and for an ordering $r_1, \ldots, r_\omega$ let $H_s = \langle r_1, \ldots, r_s \rangle \leq G$ and $I_s = [H_s : H_{s-1}]$. If $2e_s + 1 \geq I_s$ for every s, then the image of the legal exponent box $\prod [0, 2e_i]$ under $f \mapsto \prod r_i^{f_i}$ is all of H_ω, every element having a unique chain-reduced representative, and success via a target $t \Leftrightarrow t \in H_\omega$. [PROVED. The group-theoretic core is KNOWN — it is the collected normal form for polycyclic generating sequences, specialized to a finite abelian group (GAP Reference Manual §45.1; Holt–Eick–O'Brien 2005) — and its application to the legal exponent boxes of x_d^2 was not found in the Erdős–Straus literature consulted. Certificate: 12,559 of 39,405 archived composite successes, 0 false firings.]*

Theorem K5′ (rough-part failure; repaired). *Let $L \geq 3$, $p \equiv 1 \pmod 4$ prime, d admissible with every prime factor $\leq L$. Assume (H1) p is a quadratic residue modulo every odd prime $\ell \leq L$ and (H2) $p \equiv 1 \pmod 8$ or x_d odd. If the L-rough part of x_d is 1 or a prime power Q^e with e odd, then p fails at d. [PROVED. The Round 1 statement omitted (H2) and is false without it: $p = 421 \equiv 5 \pmod 8$, $d = 27$, $x_d = 112 = 2^4 \cdot 7$ succeeds in Channel I with $a = 128$ (denominators 112, 1750, 842000). Re-verified for this paper.]*

Proposition U4 (no congruence family of failures). *For every admissible d and every residue class of p (to any modulus) containing infinitely many primes ≡ 1 (mod 4), infinitely many primes in that class succeed at d — in Channel I with a = q for a prime q ≡ −4⁻¹ (mod d) dividing x_d, by Dirichlet's theorem and the Chinese remainder theorem. Hence no infinite family of failures at a fixed shift can be exhibited by congruence conditions on p alone. [PROVED, depending on the audited dependency Dirichlet.]*

Further accepted statements, listed for completeness: A4′ (witness anatomy: ε = (d/p) = (x_d/d); Channel I witnesses have χ(w) = −1; complementary witnesses only at flip shifts), A6′ (for prime d ≡ 3 (mod 4), d ∉ {3, 7, 11, 31}, the two-free-prime outcome is not a function of character values of order ≤ 6, which delimits the reach of character conditions and explains why d = 3 and 7 are complete), K1 (the exact two-defect structure — group defect and box defect — of success at composite d given local witnesses at every prime-power component), K6 canonical (for L ≥ 7; the earlier form fails at p = 15073), and U2/U3 (primitive values of a form of discriminant −d are QR-smooth modulo d; exact equivalences at d = 3 and 7 with norm forms $u^2 + uv + v^2$ and $u^2 + uv + 2v^2$).

4.3 What the criteria certify: accounting on the archive

The criteria of §4.2 were measured on the B6 residual of the W4 archive — 50,515 (p, d) rows of known success that an earlier spectral criterion missed. Figure 1 and Table 3 give the exact accounting. The units are (p, d) rows; percentages measure how much of the known success set is explained by an exact bounded-size residue test, and the charter forbids reading them as progress toward the conjecture.

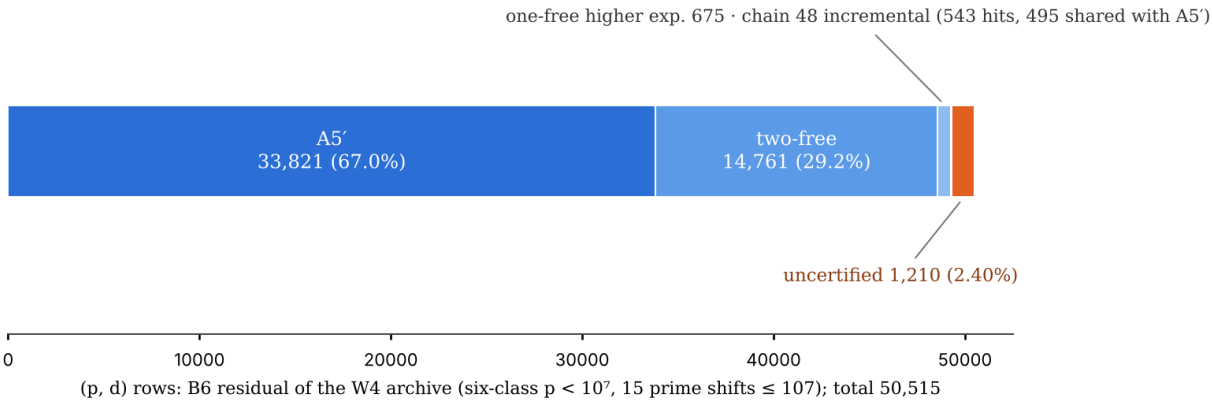


Figure 1. Certification of the archived residual successes by exact residue criteria.

Figure 1 displays the 50,515 (p, d) rows of the B6 residual of the W4 archive — six-class primes below 10^7 against the fifteen prime shifts d ≤ 107 — partitioned by the first exact criterion that certifies each, in the order A5′, two-free-prime, higher-exponent one-free, chain saturation (the chain segment is its 48 incremental rows; Table 3). A5′ alone accounts for two thirds, the two-free-prime criterion for most of the rest, and 1,210 rows (2.40%) remain, every one with at least three non-residue primes in x_d. All counts were replicated by a sealed independent validator and reproduced in GPT's review (96,257 certificates checked). The figure measures how much of the known success set an exact bounded residue test explains; the Charter forbids reading it as progress toward the conjecture.

Source: Claude Round 2, Addendum A and Round 3 records, all counts replicated by a sealed independent validator and reproduced in GPT's review (96,257 certificates checked).

Table 3. Certification accounting by criterion.

Criterion	Rows certified	Share of residual	Replication
A5′ one-free-prime residue test	33,821	66.95%	sealed validator, exact match; GPT review
Two-free-prime criterion C3-01	14,761	29.22%	preregistered count met exactly; GPT review
One-free prime, higher exponent	675	1.34%	GPT review (gain reproduced)
Chain saturation	543 hits; 48 incremental	0.10%	0 false firings on 39,405 composite successes; 495 of 543 shared with A5′ (row-level reconciliation, 30 Oct)
Certified by exact residue criteria	49,305	97.60%	—
Uncertified: 1,192 rows with three non-residue primes in x_d, 18 with four	1,210	2.40%	no pure-$\bar{p}$ criterion on the C3-01 domain; any barrier for ≥ 3 non-residue primes OPEN
Named union over the whole W4 success set, without / with the archived B6 mask	134,807 / 137,011 of 138,221	97.53% / 99.12%	unit: (p, d) rows

Table 3 gives the exact accounting behind Figure 1 in (p, d) rows, with the replication status of each count. The last row widens the population to all 138,221 archived successes: the named union of criteria certifies 134,807 of them without the archived B6 mask and 137,011 with it. The units are rows, not primes, and the percentages measure certification of already-known successes. Version 5 printed the four criterion counts as 33,821, 14,761, 675 and 543, which sum to 49,800, not to the union 49,305; GPT's review found the 495-row discrepancy. The row identifiers behind all four counts were then recovered from the sealed Round 1–3 evidence archives (NumPy membership arrays over the 307,695-row baseline; 615 manifest hashes verified) and audited by Claude: the A5′, two-free-prime and higher-exponent lists are pairwise disjoint; the chain list meets A5′ in exactly 495 rows and the other two in none; the union of the four is 49,305 rows, and its complement in the 50,515-row population is the 1,210 uncertified rows, row for row. The chain criterion therefore fires on 543 rows and certifies 48 that nothing earlier in the ordering certifies: 33,821 + 14,761 + 675 + 48 = 49,305. The reconciliation used the preserved data and required no rerun of the searches; the recovered lists accompany this version.

4.4 A conditional density theorem for failures at a fixed shift

Theorem U(a′). *For every $d \equiv 3 \pmod 4$, $d \geq 3$, squarefree or not, $\#\{p \leq N : p \equiv 1 \pmod 4, p > d, p \text{ fails at } d\} \geq c_d \cdot N/(\log N)^{3/2}$ for $N \geq N_d$, and every prime counted has*

x_d QR-smooth modulo d (the squarefree kernel of d at odd exponents). For d = 3 the bound holds within each of the six classes modulo 840; for every prime d ≠ 7 within each six class; for d = 7 with the residue of p modulo 5 prescribed. [CONDITIONAL on FHRSS, Acta Arithmetica 222 (2026), Theorem 1.1(1), arXiv:2504.20289 — a lower bound ≫ N/(log N)^{3/2} for primes p ≤ N in a progression that are primitively represented as B·f(u, v) + A for a positive definite binary quadratic form f. The arXiv statement was verified verbatim twice; the full proof is freely available on arXiv and was not re-verified within the line. Its progression clause requires the progression modulus to be coprime to 2DB, so the six-class refinement (modulus 840) does not follow by substitution and rests on separate local arguments that belong to the same audit obligation.]*

Proof idea. Apply the cited theorem to the principal form $f_0 = u^2 + uv + ((d + 1)/4)v^2$ of discriminant $-d$ with B = 4, A = −d; then $x_d = f_0(u, v)$ is a primitive value, QR-smooth modulo d by genus theory (Lemma U2), so p fails at d by Lemma U1. Finite control: for d ∈ {27, 63, 75, 99, 147, 171, 175, 243} and $p < 2 \cdot 10^6$, all 41,403 constructed (p, d) incidences (19,501 distinct primes) fail, while 8,722 of 17,923 primes with only non-primitive representations succeed — primitivity is essential. The method is that of Dahan (arXiv:2608.24035, Theorem 4.14) at shift 7; the extension to every d, the removal of the squarefree hypothesis and the six-class refinement are labelled POSSIBLE_NEW as corollaries. Together with U4 the picture at a fixed shift is complete in kind: failures have positive (log-power) density in every class, and successes are infinite in every class, so no fixed finite menu of shifts can be shown sufficient or insufficient by congruence conditions alone. Both statements concern each shift separately. That every fixed d fails on infinitely many primes does not imply that some prime fails at every d of a given menu: the intersection of the individual failure sets may be empty, and the joint arithmetic of the shifted numbers (p + d_i)/4 at one prime is a different question, left to §6.

4.5 Negative results, refutations and withdrawn claims

The protocol's value shows most clearly in what it removed. The following are REFUTED with witnesses kept, or withdrawn by their authors on review; each witness was re-verified exactly for this paper (Appendix B).

Table 4. Refuted, withdrawn and relabeled claims.

Claim as first stated	Disposition	Witness or reason
K5 (rough-part failure) without a hypothesis on p mod 8	false; repaired as K5′	p = 421 ≡ 5 (mod 8), d = 27, x_d = 112 = $2^4 \cdot 7$: Channel I fires (a = 128)
Converse of A4 (witness anatomy)	false; A4′ states the correct direction	p = 29, d = 3, x_d = 8: both channels fire
Every Type II solution at a six-class prime has a witness with a | x ("II-div universal")	REFUTED (finite, complete search justified by Lemma 2)	p = 2521 and 66529 are the only six-class p < 10^7 with Type II solutions but no a | x witness; least prime ≡ 1 (mod 4) with the property: 193
The pure Channel II part of C3-01 is the "a divides x" slice	false; withdrawn	p = 2,609,081, d = 11, x_d = 652,273 = $17^2 \cdot 37 \cdot 61$
A Lean parity hypothesis in the C3-01 template; an unconditional cardinality	defective; withdrawn	needs 4m a unit; section variables unbound

Claim as first stated	Disposition	Witness or reason
formula		
Jacobi-smoothness is a property of the whole failure set	quantifier error; corrected to the constructed subfamily	$p = 29$, $d = 7$
K6 canonical form for small L	false for $L < 7$	$p = 15{,}073$
No pure-$\bar{p}$ criterion exists for the two-free-prime family	PROVED negative (C3-01)	every QR class of $\bar{p}$ carries both outcomes on the sealed proof's domain (prime d, squarefree fibres); false at $d = 3$, where every input succeeds ($p = 1009$)
A three-/four-non-residue-prime barrier is a proved obstruction	reclassified OPEN (GPT A06)	the no-pure-numerator-residue theorem covers only squarefree two-free fibres with prime d
"$d \equiv 3 \pmod 4$ in every observed case" as an empirical finding	tautology	$d = 4x - p \equiv -p \equiv 3 \pmod 4$ identically; caught by a trial reviewer
No fixed finite list of shifts covers every qualifying prime (GPT, 29 Oct)	split: PROVED for Q-admissibility (Theorem E9); OPEN for channel coverage	for fixed d, channel coverage is a divisor condition on x_d ($d = 3$: a prime factor $\equiv 2 \bmod 3$; verified for all 273 residual primes below 100,000); the admissibility version follows from $J_d(1) = 1$ on $p \equiv 1 \pmod{\text{lcm}}$
"Energy condition covers 18 of 24 classes mod 840" (GPT, 29 Oct)	PROVED as Theorem E6(b)–(c); restates Mordell as Q-coverage	the 18 non-square classes are exactly the classically proved ones
Conjecture Q: an energy-forcing cell exists at every $p \equiv 1 \pmod{24}$ (frozen 29 Oct as ES-QP-TEST v1.0)	REFUTED (§4.8; two implementations, certificate, replication)	25 primes below 10^6; smallest $p = 66{,}361$: 54 eligible cells at 27 firing x, all $E \geq M$
Q-extended (non-residue restriction dropped)	REFUTED	$p = 944{,}329$: 129 eligible cells at 65 firing x, all $E \geq M$
Conjecture Q′ (residual form on R); target $E_\nu[M - E] \geq \varepsilon(p)$	REFUTED by consequence; target withdrawn	every cell at the failing primes has $M - E \leq 0$
Hypothesis S1: a one-prime (centered support ≤ 1) witness within the window $j \leq \min((p-1)/2, b^2)$ for every residual prime (GPT, 30 Oct)	REFUTED (§4.11; complete enumeration; GPT validator; director run pending)	$p = 1{,}977{,}361$: 40 of 441 shifts carry witnesses, none of support ≤ 1; first support-1 witness at $d = 2{,}379$; held at all 2,370 residual primes below 10^6
Every residual prime has a prime admissible shift passing the Kneser quotient budget with a target available (exploratory, GPT)	REFUTED; Proposition B remains a valid sufficient criterion	$p = 1201$: all 181 prime shifts $d < 2{,}402$ fail the budget; full divisor support exists at $d = 23$ and 31
The first six shifts with $(d/p) = -1$ always contain a firing channel (bound learned below 10^6)	REFUTED; development-range rule adopted	$p = 1{,}387{,}849$: shifts 19, 39, 47, 51, 67, 79 all fail; $d = 83$ succeeds ($(x, y, z) = (346983, 5917788136, 296344621272)$); up to eleven needed below 10^7

Table 4 is the line's negative ledger. Each row names a claim as it was first stated, its disposition, and the witness or reason that settled it; every numerical witness was re-verified by direct divisor enumeration for this paper, and the three Conjecture Q rows are treated in full in §4.8. The protocol's value shows most clearly here: these are the statements that would have stood in a record kept without blind reconstruction and independent validation.

4.6 The non-polynomial reduction and the channel-depth census

Because no polynomial identity can cover the residual classes, a proof there must choose x by a rule that is not a polynomial in p, and Proposition 2.1 says that, whatever rule chooses x, success at that x is decided by divisibility: whether $(p + d)/4$ has a prime factor in a prescribed residue class modulo d. Define the channel depth D(p) as the least admissible d at which either channel fires. Then the conjecture on the residual domain is equivalent to the *covering statement*: for every prime $p \equiv 1 \pmod{24}$ there is $d \le \Delta(p)$, $d \equiv 3 \pmod 4$, at which a channel fires, for some bound Δ. Four elementary lemmas (Appendix A.1) make this exact; in particular, for $p \equiv 1 \pmod{24}$ both channels fire at $d = 3$ exactly when $(p + 3)/4$ has a prime factor $\equiv 2 \pmod 3$, and integers free of such factors have counting function of order $X/\sqrt{(\log X)}$ (Landau), which predicts a slow decay of first-channel failures: the observed fractions 44.9% ($p \le 3 \cdot 10^6$), 42.3% ($2 \cdot 10^7$) and 40.8% ($6 \cdot 10^7$) match the predicted factor to two digits (transfer of Landau's density to shifted primes is CONDITIONAL on a Bombieri–Vinogradov-type input not re-verified).

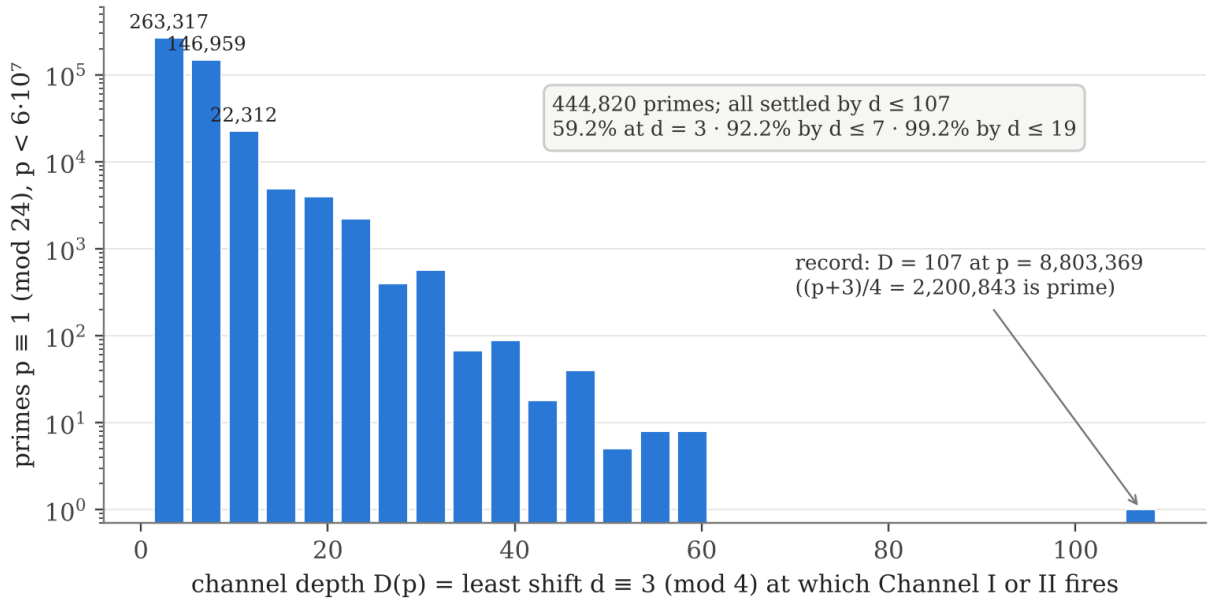


Figure 2. Distribution of the channel depth D(p) for primes $p \equiv 1 \pmod{24}$ below $6 \cdot 10^7$.

Figure 2 plots, on a logarithmic scale, how many of the 444,820 primes $p \equiv 1 \pmod{24}$ below $6 \cdot 10^7$ have each channel depth. Every prime is settled by $d \le 107$; the record is p = 8,803,369, at which $(p + 3)/4 = 2{,}200{,}843$ is itself prime. The counts fall roughly by half for each step of four in d. The census was recomputed for this paper in 22 seconds and replicated by the author (Appendix B); as the prior-art paragraph of this section explains, the statistic and its record primes were published by Ionascu and Wilson in 2011, so the figure is a replication rather than a finding.

Table 5. Channel-depth census to $6 \cdot 10^7$.

Depth bound d	Primes with D(p) = d	Cumulative	Cumulative share
3	263,317	263,317	59.20%
7	146,959	410,276	92.23%

11	22,312	432,588	97.25%
15	4,852	437,440	98.34%
19	3,971	441,411	99.23%
23	2,206	443,617	99.73%
27	396	444,013	99.82%
31	572	444,585	99.95%
39	88	444,740	99.98%
47	40	444,798	100.00%
59	8	444,819	100.00%
107	1	444,820	100.00%

Table 5 gives the cumulative counts behind Figure 2 in distinct primes: 59.2% of the residual-class primes are settled at d = 3, 92.2% by d ≤ 7, 99.2% by d ≤ 19, and all of them by d = 107. The census was recomputed on 29 October 2026 and agrees with the 28 October record and with the author's Colab run.

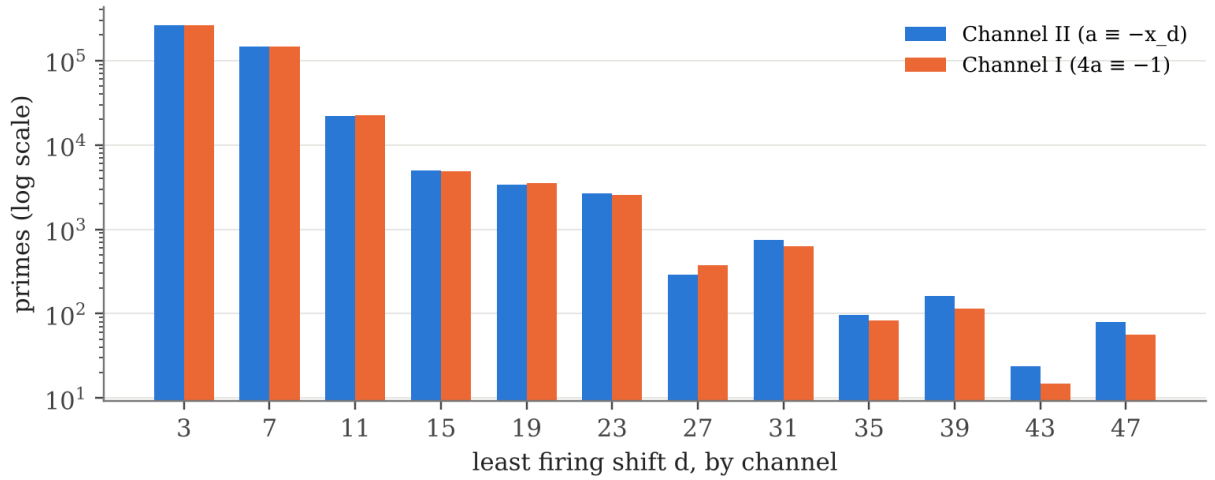


Figure 3. Least firing shift by channel.

Figure 3 separates the census by channel: for each shift d it shows how many primes first fire in Channel II (a ≡ −x_d) and how many in Channel I (4a ≡ −1). The two histograms agree to within a few percent at every d ≤ 47, so Channel I rarely decides the depth; the Type II mechanism carries the census.

Prior art (found in the literature check for this version). The channel depth is not new. Ionascu and Wilson [Ionascu–Wilson 2011; arXiv:1001.1100, 2010] define classes C_i of integers n having a solution whose first denominator is among the first i admissible values starting at ⌈n/4⌉ — for p ≡ 1 (mod 4), exactly x = (p + 3)/4, (p + 7)/4, …, so that D(p) = 4i − 1 at the least i with p ∈ C_i. They state the conjecture in the form ∪C_i = ℕ, list the primes that first require a larger class as 2, 73, 1129, 1201, 21169, 118801, 8803369 with classes C_1, C_2, C_3, C_4,

C_8, C_{15}, C_{27}, and report from computations to about $4.1 \cdot 10^9$ that no prime in a class beyond C_{27} had been seen. The record list computed for this paper over all primes $p \equiv 1 \pmod 4$ below $9 \cdot 10^6$ is 5, 73, 1129, 1201, 21169, 118801, 8803369 at depths 3, 7, 11, 23, 31, 59, 107, i.e. classes 1, 2, 3, 6, 8, 15, 27: it agrees with Ionascu–Wilson at six of seven entries, the discrepancy at $p = 1201$ was resolved for version 6: a direct factor-pair check finds no solution with first denominator 301 through 305 and one at $x = 306$ ($4/1201 = 1/306 + 1/15980 + 1/172727820$), so $D(1201) = 23$ and the class is C_6 under their own definition; the C_4 entry in the arXiv version is a misprint (the journal text was not checked). Dahan [2026] likewise reports that the smallest admissible shift never exceeds 63 for the 3,202 primes $\equiv 1 \pmod{24}$ below $3 \cdot 10^5$, consistent with the census. The census of this section is therefore labelled KNOWN (statistic) / replication (data), and the bound $D(p) \leq 107$ is reported by Ionascu–Wilson to hold to about $4.1 \cdot 10^9$, a figure this paper has not independently reproduced beyond $6 \cdot 10^7$.

This is where the honest X/Y/Z reading of the line is sharpest. X and Y are complete: the reduction is exact, the lemmas are elementary, and the census is reproducible in seconds. The Z that would be needed is a pointwise sieve result: that among roughly log p specific shifted numbers $(p + 3)/4$, $(p + 7)/4$, … at least one carries a prime factor in a prescribed class. The almost-all version is known — Vaughan (1970) bounded the exceptional set below N by N/exp(c(log N)^{2/3}) by exactly this mechanism, and Pomerance and Weingartner (2026) extend it — but sieve methods of the known kind lose a positive proportion of primes at every step, and no known sieve argument shows the exceptional set is empty. Dahan's stated exponent ceiling is likewise method-specific — it concerns his witness systems under his sieve constraint, and he leaves the extension to arbitrary Type I/II witness systems open — so neither source nor this paper shows that every change of representation must fail. The line produced no such result and claims none. Two consequences were recorded in the charter: the residual classes are not resistant for want of a trick but because the trick reduces to a sieve statement sieves cannot deliver pointwise; and no refinement at fixed p (bridge, orbit, square-class) changes this, since every such criterion is a sufficient condition at one x while the open question is existence of a good x.

4.7 The energy route: local theory, uniform families, and Conjecture Q

The forcing theorem of §4.1 suggests a sufficient condition of a different kind, developed by GPT in its final round and delivered, after its withdrawal, as a draft paper on 29 October. Claude reviewed that draft for this version: every theorem below was reconstructed from its statement and checked by an independently written program (Appendix B) — the closed-form minimum against brute-force enumeration on all 2,050 parameter cases, the classification against every admissible input at every prime $p \equiv 1 \pmod{24}$ below 1,500, the family theorems on all qualifying primes below 20,000, and the 577 and 433 examples from scratch. No mathematical error was found. The theorems are PROVED (paper proofs; author GPT; other-system review by Claude; not formalized except where stated); the density corollary is CONDITIONAL on a cited sieve bound.

Setting (frozen in ES-QP-TEST v1.0). p is prime, $p \equiv 1 \pmod{24}$. An admissible x satisfies $(p + 3)/4 \leq x \leq 3(p - 1)/4$ and $(x/p) = -1$; put $d = 4x - p$, so $d \equiv 3 \pmod 4$ and $\gcd(x, d) = 1$. H is the subgroup of $(\mathbb{Z}/d\mathbb{Z})^\times$ generated by the residues of the prime factors of x, $K = H^2$ its square subgroup, and a cell is a coset of K in H. For $r \in H$, w(r) counts the divisors of x^2 with residue r; every divisor residue lies in H. The divisor involution $a \mapsto x^2/a$ induces $\iota(r) = x^2 r^{-1}$ on residues,

with fixed point x (the central residue) and possibly others; a two-element orbit is a complement pair. The targets are t_I = −px and t_II = −x; a cell is eligible if it contains a target. For a cell C, T_C = Σ w(r) and E_C = Σ w(r)² are its mass and energy; an avoiding profile is v: C → $\mathbb{Z}_{\geq 0}$ of mass T_C with v(r) = v(ι(r)), v(x) odd and positive if x ∈ C, v even at non-central fixed residues, and v = 0 at the target; M_C is the least energy of an avoiding profile (+∞ if none). Q(p): some admissible x and eligible C have E_C < M_C.

Lemma E1 (target separation). *Each eligible cell contains exactly one target. Channel I is available (t_I ∈ H) iff −4 ∈ H; Channel II iff −1 ∈ H. [Proof: (x/p) = (d/p) = J_d(p) = −1 by reciprocity; p ≡ 4x gives J_d(x) = −1, and d ≡ 3 (mod 4) gives J_d(−1) = −1, so J_d(t_I) = −1 and J_d(t_II) = +1 while the Jacobi symbol is constant on cosets of K.]*

Theorem E2 (forcing and reconstruction). *If E_C < M_C on an eligible cell, a divisor a of x² occupies the target and yields a positive solution of 4/p: in Channel I, y = (px + a)/d, z = (px + p²·x²/a)/d; in Channel II, y = p(x + a)/d, z = p(x + x²/a)/d. If Q(p) holds for every prime p ≡ 1 (mod 24), the conjecture holds for every n ≥ 2 (the classes p ≡ 2 mod 3, p ≡ 3 mod 4 and p ≡ 13 mod 24 are disposed of by the identities (p, (p+1)/3, p(p+1)/3), ((p+1)/4, 2p(p+1)/4, 2p(p+1)/4) and the d = 3, a = 2 witness). [Verified on every odd prime below 5,000.]*

Theorem E3 (exact avoidance minimum). *Remove the involution orbit of the target; let b be the number of surviving complement pairs, f the surviving non-central fixed bins, c ∈ {0, 1} the surviving central bin, D = 2b + f + c, T = T_C. If the central bin was removed, M = +∞. Otherwise feasibility requires T = c + 2U with U ≥ 0 an integer; if D = 0 then M = 0 for T = 0 and +∞ for T > 0; if D > 0, writing U = qD + r (0 ≤ r < D), r_1 = min(r, b), r_2 = min(max(r − b, 0), f), r_3 = max(r − b − f, 0): M = c + 4Dq² + 4cq + (8q + 2)r_1 + (8q + 4)r_2 + (8q + 6)r_3. Consequently M = T iff U ≤ b, so finite forcing needs T > 2b + c; E, M and T have the same parity and every positive finite margin M − E is an even integer ≥ 2; and T²/D ≤ M ≤ T²/D + D − c. [Proof: an exchange argument on the increasing marginal costs 4k + 2 (pair), 8k + 4 (fixed bin), 8k + 8 (central) per unit of two in mass; Appendix A.4. Checked against brute-force minimization on all 2,050 cases b, f ≤ 4, c ∈ {0,1}, T ≤ 40: 0 mismatches.]*

Proposition E4 (collision form). *With N_C the number of unordered pairs of distinct divisors of x² sharing a bin of C, E = T + 2N_C; for finite M, forcing is equivalent to N_C < L = (M − T)/2 = Dq(2q − 1) + 2cq + 4qr + r_2 + 2r_3. A sufficient continuous test is D·E < T²; it is not necessary, so a failed variance test refutes nothing. Over all cells, N = Σ τ(x²/(uv)) over coprime u < v with uv | x² and d | (v − u). [Verified on the same 2,050 cases.]*

Proposition E5 (support obstruction). *If a cell forces with finite M, every complement pair of the original cell is occupied; hence T ≥ 2B + c (B the pairs of the whole cell), and an argument from small collision energy alone is incomplete.*

Theorem E6 (uniform families; restate classical coverage). *(a) If p + 1 has a prime factor ≡ 3 (mod 4), then for any odd d | p + 1 with d ≡ 3 (mod 4), x = (p + d)/4 is admissible and the Channel I target is central, so M = +∞ and Q(p) holds; conversely a central target exists only when d | p + 1. The solution is x, y = x(p + 1)/d, z = py (Obláth's family). (b) If p mod 7 ∈ {3, 5, 6}, Q(p) holds at d = 7: the target is central*

when $p \equiv 6$, and when $p \equiv 3$ or 5 it lies in the complementary pair of the central cell $\{x, 2x, 4x\}$, giving a finite margin $M - E = 4ak + 2k^2 \geq 6$. (c) If $p \bmod 5 \in \{2, 3\}$, $Q(p)$ holds at $d = 15$ with the Channel I cell $\{11, 14\}$ consisting of the target orbit alone and carrying positive mass. Together (b) and (c) cover the 18 non-square reduced classes modulo 840 — Mordell's classical coverage restated as Q-coverage, which is what GPT's undelivered '18 of 24' claim meant. [(a) verified on all admissible inputs below 2,000; (b), (c) on all qualifying primes below 20,000: 0 violations.]

Corollary E7 (density; CONDITIONAL). *$Q(p)$ holds for a relative-density-one subset of primes $p \equiv 1 \pmod{24}$: the possible failures lie among primes p for which $p + 1$ has no prime factor $\equiv 3 \pmod 4$, whose number up to N is O(N/(log N)^{3/2}) (the bound Pomerance and Weingartner record as a consequence of Obláth's criterion), against $\pi(N; 24, 1) \sim N/(8 \log N)$. CONDITIONAL on that sieve bound, cited and not re-verified; it gives no cutoff and the exceptional set may be infinite.*

Theorem E8 (complete classification of infinite thresholds). *For an eligible cell of an actual divisor profile at $p \equiv 1 \pmod{24}$ and admissible x, $M_C = +\infty$ if and only if: Channel I with a central target $(d \mid p + 1)$; or Channel I with $d = 15$ and $p \bmod 5 \in \{2, 3\}$; or Channel II with $K = \{1\}$ and $-1 \in H$. [Proof idea: a non-central infinite threshold needs the target orbit to exhaust the cell, so $|K| \leq 2$; in Channel I, $16 \in K$ forces $d \mid 255$, and parity, $p \equiv 1 \pmod 3$ and $J_d(x) = -1$ eliminate $d = 3, 51, 255$; in Channel II the target $-x$ is a non-central fixed residue and $K = \{1\}$ makes H an elementary 2-group whose every element is a squarefree divisor of x. Verified exhaustively on every eligible cell of every admissible input at every $p \equiv 1 \pmod{24}$ below 1,500: 16 central, 13 $d = 15$ and 1 Channel-II instance, 0 violations. Example of the third mechanism: $p = 433$, $x = 113$, $d = 19$, $H = \{1, 18\}$, $K = \{1\}$, target 1 occupied by the divisors 1 and 113^2.]*

Theorem E9 (no finite menu of shifts; correctly scoped). *No fixed finite set D_0 of shifts $d \equiv 3 \pmod 4$ supplies an admissible $x = (p + d)/4$ for every prime $p \equiv 1 \pmod{24}$: for $p \equiv 1 \pmod{\mathrm{lcm}(24, D_0)}$, which Dirichlet supplies infinitely often, $(x/p) = (d/p) = J_d(p) = J_d(1) = +1$ for every $d \in D_0$, so no x is admissible. [Depends on the audited dependency Dirichlet. Verified on 13 primes $\equiv 1 \pmod{175{,}560}$ for $D_0 = \{3, 7, 11, 15, 19\}$.] This is the correct form of the 'finite-list obstruction' that version 1 relabeled OPEN: it concerns the non-residue admissibility restriction of the Q specification only. The channel-covering version — whether some finite menu of shifts has a firing channel at every residual prime — remains OPEN, and the census of §4.6 shows no counterexample below $6 \cdot 10^7$ (nor, by Ionascu–Wilson, below $4.1 \cdot 10^9$).*

Proposition E10 (refutation certificate). *$Q(p)$ is refuted at p by a valid avoiding profile of energy $\leq E_C$ for every admissible x and eligible cell C, and conversely; a failed cell, a failed x or an interrupted computation refutes nothing, and a refutation of Q limits the route, not the conjecture.*

Let R be the set of primes in the six square classes modulo 840 all of whose odd prime factors of $p + 1$ are $\equiv 1 \pmod 4$. Theorems E6 and E8 reduce universal Q to the residual form Q′: for every $p \in R$ there is an admissible x for which either Channel II has $K = \{1\}$ and $-1 \in H$, or some eligible cell has finite M_C and $N_C < (M_C - T_C)/2$. $Q' \Leftrightarrow Q$ after the proved reductions, and

Q′ ⇒ ES by Theorem E2. Both were stated as conjectures in GPT's draft of 29 October; §4.8 refutes them.

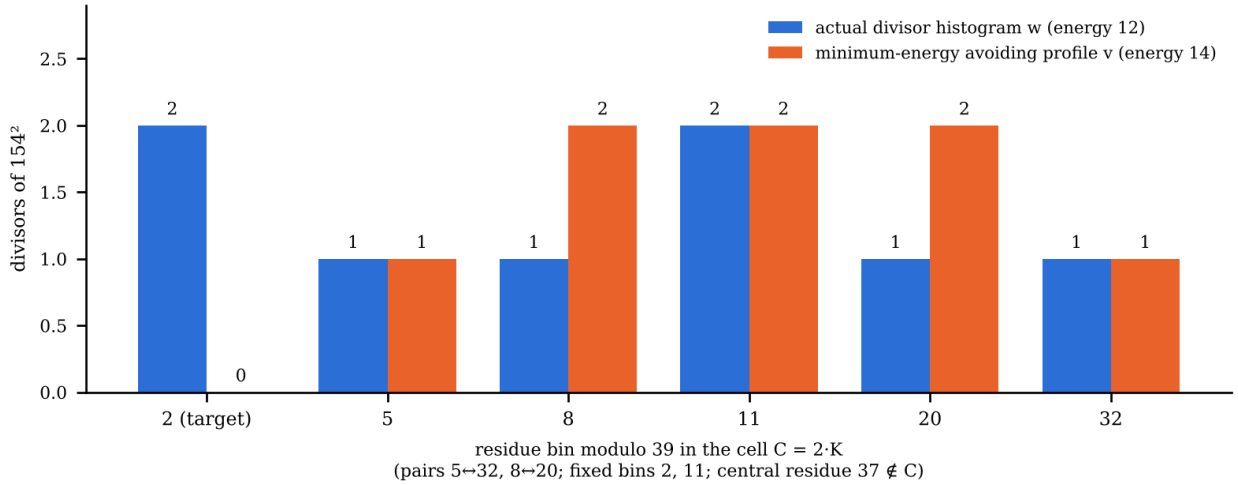


Figure 4. The certified energy-forcing instance at p = 577.

Figure 4 shows the one instance of energy forcing that is Lean-certified. At x = 154, d = 39, Channel II, the actual divisor histogram on the cell {2, 5, 8, 11, 20, 32} has mass T = 8 and energy E = 12 (blue). Removing the target orbit {2} leaves two complement pairs (5 ↔ 32, 8 ↔ 20), one fixed bin (11) and no central bin, so Theorem E3 gives the avoidance minimum M = 2·2 + 4·1 + 6·1 = 14, attained by the orange profile. Since 12 < 14 the target bin must be occupied; the forced divisor is a = 2, giving 4/577 = 1/154 + 1/2308 + 1/177716. Both values were recomputed for this paper. The figure was redrawn for version 6.2 so that its only title is this caption.

Pilot (FINITE_VERIFIED, 29 Oct). Two sealed implementations written from the frozen specification without sight of each other — Claude's marginal-cost checker (29 primes to 1873 within the 5 CPU-minute cap) and GPT's dynamic-programming implementation GPT-DP-1.0.0 (37 primes to 2377) — agree on every mathematical field for the 29 common primes (10,884 eligible cells; 10,844 finite thresholds, 40 infinite, 161 witnesses), and Q(p) holds at each. The pilot range lies below the smallest counterexample of §4.8.

4.8 Refutation of Conjecture Q and of its extension

On the evening of 29 October the author asked for a computational attack on Q′, suggesting a Monte Carlo approach. The decisive observation is that no sampling is needed. A counterexample to Q at p requires every admissible x to fail, about p/4 inputs; but E_C < M_C implies the target bin is occupied (Theorem E2), an occupied target is a firing channel, channel firing is decided cheaply by divisor residues (Proposition 2.1), and only a few dozen x per prime fire. So Q is decided exactly by scanning all admissible x, computing cells only at firing x, and applying Theorem E3 for M_C — seconds per prime below 10^6. The program (q_attack.py, standard-library Python, exact integers, written from the statements of §4.7 without sight of GPT's code) was calibrated before use: it reproduces the pilot record exactly (10,884 / 40 / 161; Q at all 29 primes) and reproduces mass, energy, minimum, force verdict and the orders of H and

H^2 on all 1,979 eligible cells of GPT's sealed Monte Carlo experiment (§4.9) with zero mismatches.

Theorem Q1 (refutation of Q). *Among the 9,732 primes $p \equiv 1 \pmod{24}$ below 10^6, exactly 25 admit no admissible x and eligible cell with $E_C < M_C$. The smallest is $p = 66{,}361$. All 25 lie in R; they are 25 of the 1,494 R-primes below 10^6 (1.67%) and occur in every band of 10^5. [REFUTED label for Q. Finder: Claude, 29–30 Oct. Confirmation at 66,361: GPT, 30 Oct (§4.9). Replication: the author, Colab, 30 Oct 10:41 ET, output identical.]*

Table 6. The 25 counterexamples to Conjecture Q below 10^6.

p	p mod 840	firing x	eligible cells at firing x	least firing d	cells with E = M = T
66,361	1	27	54	31	30
155,521	121	43	86	47	48
176,089	529	38	76	59	55
196,561	1	33	66	59	38
309,241	121	50	99	79	59
322,921	361	29	58	47	28
386,401	1	34	68	59	43
393,121	1	31	62	71	37
411,049	289	36	72	59	45
537,769	169	50	100	31	74
545,449	289	55	110	39	68
629,281	121	56	110	47	69
670,849	529	62	124	51	81
685,609	169	65	130	19	88
762,721	1	53	106	39	66
765,241	1	45	90	31	48
789,721	121	63	125	31	70
810,049	289	56	112	39	76
852,889	289	59	118	31	75
869,521	121	59	118	59	76
936,769	169	62	124	31	71
944,329	169	45	89	47	54
946,969	289	50	100	31	60
950,161	121	81	161	43	112
991,201	1	53	106	55	56

Table 6 lists the 25 primes at which Conjecture Q fails, with the residue class modulo 840, the number of admissible x at which a channel fires, the number of eligible cells at those x, the least firing shift, and the number of cells in the collision-free regime E = M = T. At every eligible cell at every firing x the energy satisfies E ≥ M, so no cell forces. The last column is the diagnosis: in the collision-free regime Corollary E6 (M = T exactly when U ≤ b) makes forcing impossible in principle, and most of the cells at every failing prime lie there. The full cell lists (d, x, channel, T, E, M) for all 25 primes are in the deposited record q_attack_results_1M_claude.json (SHA-256 f5744897…2ba21), the data of a Proposition E10 certificate.

Example, p = 66,361 = 840·79 + 1, p + 1 = 2·33181 (33181 prime, ≡ 1 mod 4), so p ∈ R. There are 16,658 admissible x; 27 fire a channel, giving 54 eligible cells; 32 have E = M and 22 have E > M. The first three firing inputs: d = 31 (x = 16,598), Channel I T = E = M = 9, Channel II T = 18, E = 28, M = 26; d = 43 (x = 16,601), T = E = M = 3 and T = 6, E = 8, M = 6; d = 47 (x = 16,602), T = E = M = 9 and T = 18, E = 24, M = 18. Erdős–Straus holds in two ways worth distinguishing: at the admissible x = 16,598, 4/66361 = 1/16598 + 1/51363414 + 1/115269057 (GPT's example — a solution inside Q's own domain that no energy cell detects, but that Theorem A5′ certifies: see below), and at the residue x = 16,591, excluded by the specification, 4/66361 = 1/16591 + 1/368038106 + 1/129917451418.

The specification admits only x with (x/p) = −1, a restriction inherited from the H_N formulation; Lemma E1 uses it, but the forcing implication of Theorem E2 does not. Define Q-extended by dropping it. Below 10^6, Q-extended holds at 9,731 of 9,732 primes, and at 6,840 of them the first forcing x is a quadratic residue — the restriction discards most of the cheap witnesses, including all of them at 24 of the 25 failing primes. It does not save the route.

> **Theorem Q2 (refutation of Q-extended).** *At p = 944,329 (≡ 169 mod 840; p + 1 = 2·5·94433) exactly 65 values of x in (p/4, 3p/4), of either residuosity, fire a channel, giving 129 eligible cells; at every one E ≥ M. No energy comparison of the form of §4.7 certifies a solution at this prime, although 4/944329 = 1/236094 + 1/4772638766 + 1/780326438241 (Channel II at d = 47). [REFUTED. Claude; author's replication.] Counting convention (added in version 6): 129 is the number of channel-labelled eligible-cell incidences. With the non-residue restriction dropped, Lemma E1 no longer separates the targets, and at 16 of the 65 firing x both targets lie in one cell, so there are 113 distinct cells. Avoiding each target separately and avoiding both at once are different specifications; GPT's review computation of 30 October (§4.9) checked the stronger simultaneous condition on all 16 shared cells and found an explicit avoiding allocation of energy at most E in each, so Q2 stands under either convention. The witness has small support: at d = 47, x = 236,094 = 2·3·19²·109 and the Channel II divisor is a = 1,444 = 2²·19².*

> **Corollary Q3.** *Conjecture Q′ is false, and the proposed research target — an arithmetic distribution ν_p on admissible cells with E_ν[M_C − E_C] ≥ ε(p) > 0 for every residual prime — is unattainable as stated, since at the failing primes every cell has M_C − E_C ≤ 0.*

What survives, and the diagnosis. Every theorem of §4.7 stands; none asserted Q. The energy test remains a correct sufficient condition certifying all primes outside R (Theorem E6) and 98.3% of R-primes below 10^6, with 94% of first witnesses at d ≤ 23. At a failing prime the firing cells are in or near the collision-free regime: occupancies 0 or 1, E = T, and T ≤ 2b + c so that M

= T too. The histogram carries no excess energy, so no comparison of occupancies can force anything, although the solutions exist. This places the energy route near where the residue criteria of §4.2 and the sieve reduction of §4.6 stand: an exact, bounded method that stops on inputs whose shifted numbers (p + d)/4 have few, well-spread divisors. Version 5 went further and asserted that the three methods stop on the same primes; GPT's review showed that this is false at the smallest counterexample itself. At p = 66,361 the first Q-admissible firing input is d = 31, x = 16,598 = 2·43·193, whose prime factors have Legendre symbols (+1, −1, +1) modulo 31; it is therefore an A5′ input with m = 386 and s = 43, and A5′ certifies Channel II there through the divisor a = 193² = 37,249 ≡ 18 ≡ −x (mod 31) — exactly the identity 4/66361 = 1/16598 + 1/51363414 + 1/115269057 displayed above (p ≡ 21 ≡ −4a (mod 31), as the criterion requires). The energy test discards that witness; the residue test keeps it. Since all three methods read the same divisor residues, they are related representations of one divisor condition rather than independent measurements of one barrier, and the correct diagnostic is a method-by-input table — for each (p, d): whether A5′, C3-01, chain saturation and energy forcing certify it, what divisor witness exists, and what information each failing criterion discarded. That table is listed among the next steps. The question the line is left with is unchanged: what certifies a firing channel when the divisor structure of x_d is sparse.

4.9 Confirmation, cross-validation and public deposit

GPT's confirmation (ES-Q-REF-1, 30 Oct 11:17 ET). On a task order relayed by the author, GPT reran its unchanged sealed GPT-DP-1.0.0 core at p = 66,361 — eleven sealed files hash-verified before and after, a separately identified adapter (ES-Q-REF-1-ADAPTER-1.0.0) lifting the archived frontend's p ≤ 13,000 bound without touching sealed sources — and obtained 16,658 admissible x, 27 firing, 54 eligible cells, 32 with E = M, 22 with E > M, none with E < M, matching the task-order table in all 54 rows. A second GPT program importing no sealed code validated the complete Proposition-E10 certificate: 54 explicit avoiding allocations (each checked for cell membership, exact mass, pair symmetry, central and fixed-bin parities, zero at the target, and energy ≤ E_C) plus the actual-profile rule for the 16,631 non-firing inputs. The certificate was produced before comparison with the table; total measured CPU 5.23 s. Claude then audited GPT's evidence package independently: manifest hashes 15/15; admissible and firing sets re-derived and identical; all 54 allocations re-validated by Claude's own code (54/54), with T, E, M agreeing on every row. GPT scoped its confirmation honestly — the counterexample itself, not minimality, not the other 24, not Q-extended — and this paper keeps that distinction. GPT's review of version 5 (30 October) added a second, differently constructed check: a new producer written from the statements of §4.7 without inspecting the project implementation, hashed before execution, and a validator importing no producer or optimizer code that re-factored inputs, rebuilt the groups and re-enumerated the divisor data. It re-decided Q at 66,361 (16,658 admissible x; 27 firing; 54 incidences; E = M in 32, E > M in 22, E < M in none) and Q-extended at 944,329 (472,164 integers; 65 firing; 129 incidences on 113 distinct cells; 85 / 44 / 0), validated every per-target allocation (54 of 54; 129 of 129) and the 16 shared cells under simultaneous avoidance, in 34.25 CPU-seconds of completed runs. The manuscript counts were visible to it, so this is implementation independence within GPT, not blind rediscovery; its records carry their own version and receipts in the review evidence archive.

GPT's Monte Carlo experiment (ES-MC-1.0.0, 29 Oct). Independently of the refutation, GPT ran a preregistered sampling experiment on 34 primes in the six classes up to 45,361 (1,224 draws under three selection rules: uniform, divisor-rich, small d), which the author reproduced

byte-for-byte on Colab. Its findings are consistent with the exhaustive picture: uniform sampling found no forcing in 408 draws, small-d sampling found it in about 10% of draws, and 1,823 of its 1,916 finite cells sat at the baseline M = T. The experiment could not see the failures — all its primes lie below 66,361 — but its 1,979 recorded cells were the cross-implementation check that validated Claude's program before the sweep; the two agree on every cell.

Author's replications. Census (29 Oct 18:18 ET, 34.3 s): identical. Energy-route checker (29 Oct 18:55 ET): identical. Q sweep to 120,000 (30 Oct 10:05 ET) and to 1,000,000 (30 Oct 10:41 ET, 876.7 s): identical to the last digit, including the 25 primes, the single Q-extended failure and the first-witness distribution.

Public deposit. On 30 October the author deposited every jointly validated artefact in the public repository github.com/whouxe82/es-lean-certification, folder es-xyz-validated/ (commit ff9451d; README pointer in commit aa3c4ba): the census program and $6 \cdot 10^7$ output; the energy-route theorem checker; the Q-decision programs with the 300,000 and 10^6 records; the author's Colab output; GPT's ES-Q-REF-1 report and evidence package (which contains the unchanged sealed GPT-DP-1.0.0) with Claude's validator; GPT's ES-MC-1.0.0 results, the author's reproduction receipt and Claude's cross-check; the pilot calibration; a STATUS file giving the label of every result; and SHA-256 sums. Claude verified the deposit against its manifest from an anonymous clone. The Lean certification package of §4.1 is the repository's root. Not deposited: the Round 1–3 sealed archives behind §4.2 (cross-reviewed but not director-replicated; their row-level contents were recovered and audited for version 6, §4.3, and their deposit awaits the author's decision). The paper itself was not part of the 30 October deposit; the PDF of this version is added to the same repository at paper/Houze_ES_Conjecture_v6.5.pdf, with its SHA-256 hash recorded in the repository README, so that the SSRN copy and the repository copy can be confirmed identical byte for byte.

GPT's final 29 October report, once delivered as a draft paper, was reviewed in full for version 2: its six reported items resolved as Theorem E6(b)–(c) (the '18 of 24' statement, restating classical coverage), E6(a) and E8 (the (p + 1) family, Obláth), E3 (the threshold formula), the 2,050-case check (reproduced), E7 (CONDITIONAL on a cited sieve bound) and E9 (the finite-menu theorem, correct in its scoped form; the channel-coverage form remains OPEN). Nothing in that draft was false; one presentation defect (no title) is noted for the record; its citation of the 10^{18} verification as 'Mihnea and Bogdan' may be correct — the second author's surname is unsettled between sources (see the reference list).

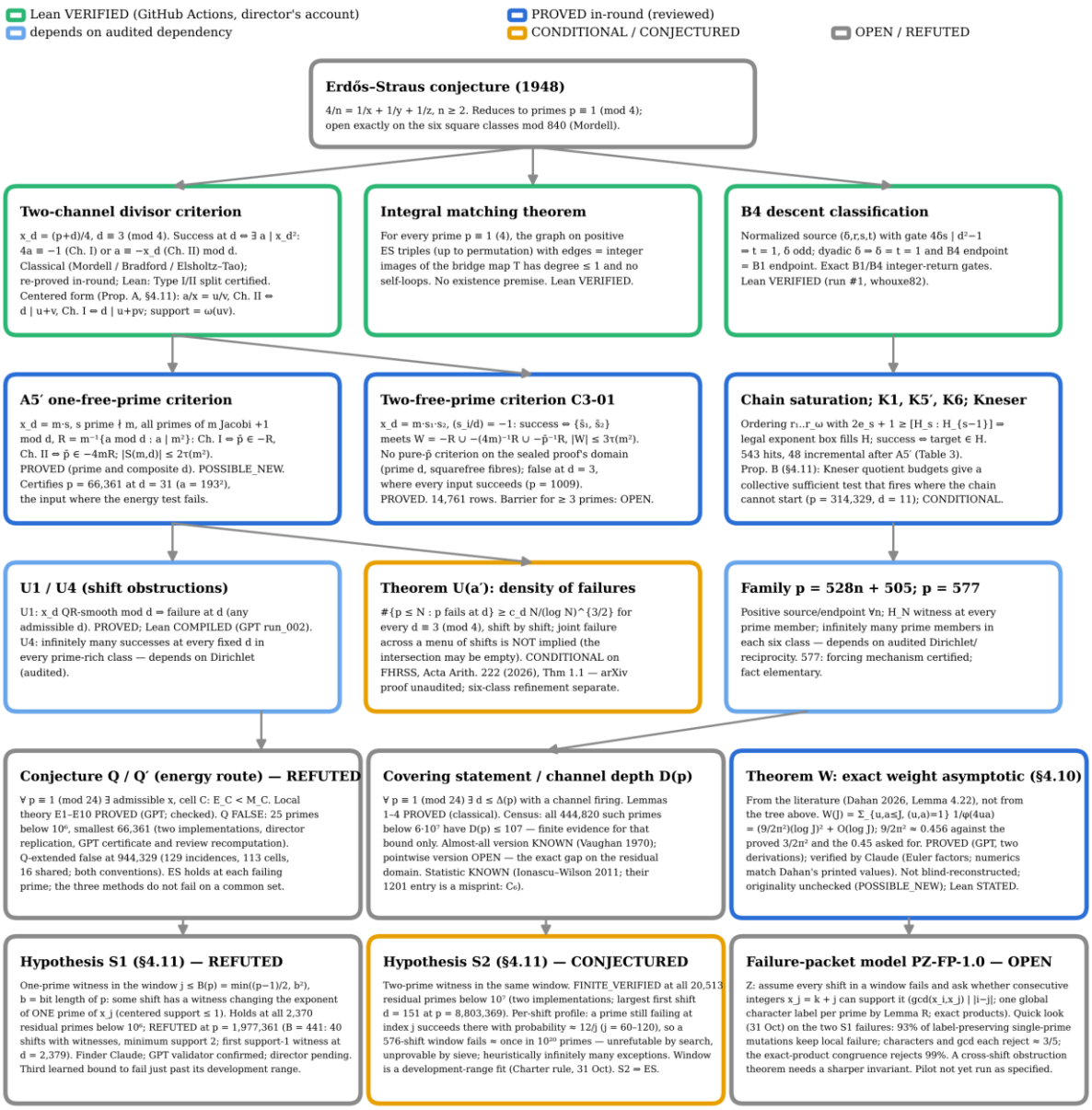


Figure 5. Proof-status map of the ES XYZ line, 31 October 2026 (drawn for version 6.3; unchanged in 6.5).

Figure 5 maps every result of the line to its evidence label by border color: green for Lean-VERIFIED statements rebuilt on the author's account, dark blue for theorems PROVED and reviewed within the line, light blue for results that depend on an audited dependency (Dirichlet, reciprocity), amber for CONDITIONAL or CONJECTURED statements, and grey for what is OPEN or, in the case of Conjecture Q, REFUTED. Arrows indicate logical dependence. Read from top to bottom the map gives the shape of the paper: the classical criterion and the certified structural theorems at the top, the exact residue criteria in the middle, the shift obstructions and the density theorem below them, and at the base the two statements on which the line stopped — the refuted energy conjecture and the open covering statement — together with the one result that entered from outside the tree, the exact weight asymptotic of §4.10, drawn with its proved-

and-reviewed border and its remaining obligations (blind reconstruction, originality) written into the box. The map was redrawn for version 6: the A5′ bound now reads $2\tau(m^2)$, the C3-01 box carries the domain of its impossibility clause, the chain box the reconciled 48, and U(a′) the per-shift qualification. Version 6.3 adds a fifth row for the Protean Z round: Hypothesis S1 (REFUTED), Hypothesis S2 (CONJECTURED, finite-verified to 10^7) and the failure-packet model (OPEN), with the centered form of the divisor criterion noted in the top-left box and the Kneser budget in the chain box.

4.10 An exact asymptotic for a published divisor weight (added in version 6)

This result arose in GPT's review of version 5 and is reported here under the label it has earned, which is not yet the label the protocol requires. Dahan's Lemma 4.22 defines $W(J) = \Sigma\, 1/\varphi(4ua)$ over ordered pairs of positive integers u, a ≤ J with gcd(u, a) = 1, with no parity or ordering restriction, and proves $W(J) \geq (\kappa_1 + o(1))(\log J)^2$ with $\kappa_1 = 3/2\pi^2 \approx 0.152$, obtained from the crude bound $\varphi(m) \leq m$; the leading factor enters his depth-density trade-off. His concluding questions record that the computed ratios $W(J)/(\log J)^2$ at $J = 10^2, 3\cdot10^2, 10^3$ are 0.62795, 0.59503, 0.57114, that the true limit appears to be close to 0.46, and ask for a proof that the constant is at least 0.45. The XYZ reading is exact: X is the lemma as stated; Y is that the lower estimate discards the prime-by-prime arithmetic of u and a, so that no finite computation can fix the constant or its error; Z is to keep that arithmetic by writing the pairwise sum as a two-variable multiplicative convolution and evaluating its Euler product.

> **Theorem W (exact weight asymptotic).** *With W(J) as above and natural logarithms, $W(J) = (9/2\pi^2)(\log J)^2 + O(\log J)$ as $J \to \infty$; $9/2\pi^2 \approx 0.4559453264$, three times the constant of the cited lemma and above the 0.45 asked for. [PROVED (paper proof, Appendix A.7): GPT, two derivations on 30 October, the second reconstructed from the statement alone before sight of the first, with a separate error-term audit and exact rational agreement of the direct and convolution forms for J = 1, ..., 32. Claude verified the coefficient by a different route — the local Euler factors of the two-variable coprime Dirichlet series, $1 + q^{-2}$ at odd q and 3/4 at q = 2, with the overall factor ½ — checked the local polynomials and the error argument of the proof line by line, and computed W(J) directly to J = 3,000: the ratios at $J = 10^2, 3\cdot10^2, 10^3$ reproduce Dahan's printed values to five digits, and the residual $(W(J) - (9/2\pi^2)(\log J)^2)/\log J$ is stable at 0.79–0.80 from J = 100 to 3,000, the signature of a correct leading coefficient with a secondary term of order log J. Not a sealed blind reconstruction: the proof reached Claude in the same packet as the statement card, and the record says so. Novelty POSSIBLE_NEW pending the citation-tool search for a prior statement of this constant; Lean STATED only.]*

What the theorem does and does not do. It answers the question Dahan poses about his own lemma, by an exact asymptotic rather than the lower bound he suggests. It is elementary — a routine Euler-product computation for an analytic number theorist — and no priority is claimed beyond having carried it out for this weight. It does not change the exceptional set, remove it, or alter the exponent ceiling of the cited method; where the improved constant enters that paper's later results, and whether it sharpens any explicitly cited bound, is a separate audit not performed here, and no sharper exceptional-set statement is asserted. The proposed next computation is a sealed-implementation pilot comparing reduced rational values of the direct and convolution forms for every J from 1 to 128, with the residual diagnostic above reported as a labelled floating-point quantity; the pilot checks the implementations, and the proof carries the theorem.

4.11 The Protean Z round: simple witnesses, a collective criterion, and the failure-packet model (added in version 6.3)

On 30 October the author asked both systems for a "Protean Z Wild Card": a deliberate change of representation rather than a further refinement of the criteria above. GPT's response, delivered as a report, a statement-only card and an evidence archive, produced four exact statements, two universal hypotheses and two refuted proposals; Claude replicated its computation, extended it by an order of magnitude with an independent implementation written from the statement card, and refuted one of the hypotheses. Everything in this section carries the labels it earned by 31 October; nothing in it bears on the conjecture itself.

> **Proposition A (centered coprime witness; classical).** *Every divisor a of x^2 has a unique reduced ratio $a/x = u/v$ with $\gcd(u, v) = 1$, $u \mid x$ and $v \mid x$, and conversely; write $x = muv$. Channel II fires exactly when $d \mid u + v$ and Channel I exactly when $d \mid u + pv$, with return maps $(x, y, z) = (muv, pmuh, pmvh)$, $h = (u + v)/d$, and $(muv, muh, pmvh)$, $h = (u + pv)/d$. Define the centered support of a witness as $\omega(uv)$, the number of distinct primes whose exponents differ between a and x. [PROVED; classical divisor algebra. At parameter level the Channel II form is the Type II family of Dahan's Theorem 3.9 with $(u, a, v, \alpha, c) = (u, h, v, m, d)$; his depth bound and the support bound below are different restrictions. The median divisor $a = x$ (support 0) is Obláth's criterion $d \mid p + 1$; the extremes $a = 1$ and $a = x^2$ give one unordered Channel II solution; no square divisor can serve Channel I, since its target $-px \equiv -4x^2$ has Jacobi symbol -1.]*

> **Hypotheses S1 and S2 (GPT, 30 October).** *For a residual prime p put $k = (p - 1)/4$, b the bit length of p, and $B(p) = \min((p - 1)/2, b^2)$. S1: some index $1 \le j \le B(p)$ has a witness at $x_j = k + j$, $d_j = 4j - 1$, of centered support at most 1. S2: the same with support at most 2. S1 $\Rightarrow$ S2 $\Rightarrow$ ES on the residual classes. [S1: REFUTED. S2: FINITE_VERIFIED below 10^7, CONJECTURED universally; its window was fitted to the development range and is a CANDIDATE bound under the Charter's development-range rule.]*

What was computed. GPT's producer and a separate GPT validator covered all 2,370 residual primes below 10^6 and the stress prime 8,803,369 (4,742 certificates). Claude replicated that run exactly, then wrote an independent checker from the statement card and ran it to 10^7: 20,513 residual primes, two certificates each, all 41,025 re-verified by the identity $4xyz = p(xy + xz + yz)$ after reconstructing y and z from the return map. GPT's validator re-derived every first shift and found no disagreement. The exchange was statement-first in the sense that Claude's implementation saw neither GPT's code nor its rows; it was not a sealed blind reconstruction of the propositions.

> **Theorem S1′ (refutation of S1).** *At $p = 1{,}977{,}361$ ($\equiv 1 \bmod 840$; $B = 441$, largest allowed shift $d = 1{,}763$) no witness of centered support at most 1 exists at any index in the window. Of the 441 shifts, 40 carry witnesses; the minimum centered support over the window is 2, attained already at $d = 7$ ($u = 2$, $v = 19$, Channel II); the first support-1 witness lies at $d = 2{,}379$, index 595, with $4/1977361 = 1/494935 + 1/411376700 + 1/145605803477077300$ (Channel I, $u = 1$, $v = 179$). [REFUTED, by complete enumeration of every divisor of x_j^2 for every $j \le 441$ (Claude, 30 October; certificate deposited with the day archive); confirmed by GPT's separate validator (33,249 divisors examined). S1 holds at every residual prime below 10^6; this is the third universal*

hypothesis of the line — after Q and the six-shift policy below — to fail immediately beyond the range on which its bound was fitted.]

S2 survives: the largest first shift below 10^7 is d = 151, at the published depth record p = 8,803,369, against a window of 576; below 10^6 it is 59. The stress prime shows the trade S2 is about: its earliest solution of any kind, at d = 107, needs three adjusted primes; a two-prime witness appears at d = 151 and a one-prime witness at d = 195. Within the window, simplicity is bought with depth.

Proposition B (Kneser quotient budget; GPT). *Write $x = \prod r_i^{e_i}$, let $H \le (\mathbb{Z}/d\mathbb{Z})^x$ be generated by the r_i, and $S = \prod_i \{r_i^k : -e_i \le k \le e_i\}$ the centered product set. If for every proper subgroup $K < H$, with $o_i(K)$ the order of r_iK in H/K, $\Sigma_i (\min(2e_i + 1, o_i(K)) - 1) \ge |H/K| - 1$, then $S = H$, and a channel fires whenever -1 or $-p$ lies in H. For prime d and H cyclic of order h it suffices to check each divisor $q > 1$ of h with $o_i(K) = o_i/\gcd(o_i, h/q)$. [PROVED from Kneser's theorem on the stabilizer of S; CONDITIONAL on Kneser under the Charter's dependency rule until a declaration in the pinned Mathlib is identified (a formalization exists outside Mathlib). Claude checked the proof and the example p = 314,329, d = 11, $x = 78{,}585 = 3\cdot5\cdot13^2\cdot31$: factor orders 5, 5, 10, 5 against segment lengths 3, 3, 5, 3, so chain saturation cannot start in any ordering, while the quotient budgets $1 \ge 1$, $10 \ge 4$, $10 \ge 9$ pass and all ten residues are reached. An exact improvement over chain saturation at that input; no new coverage claimed; originality not assessed.]*

Proposition C (compatibility between shifts) and Corollary D. *With $x_j = k + j$, $\gcd(x_i, x_j)$ divides $|i - j|$, so a prime larger than R divides at most one of R consecutive shifted integers; and by Lemma R every prime factor r of x_j has $(r/d_j) = (r/p)$, one global character label per prime across all rows. Corollary D: no finite decision tree whose tests are fixed congruences and thresholds and whose leaves return fixed rational polynomial triples can solve ES for every prime (choose a modulus L absorbing the tests; all large primes $\equiv 1 \bmod L$ take one branch; Dirichlet and the square-class obstruction finish). [PROVED; elementary. D does not exclude a fixed menu of shifts tested by factorization.]*

Two further proposals failed and are kept. The universal form of Proposition B — that every residual prime has some prime admissible shift passing the budget with a target available — is false at p = 1201: all 181 prime shifts below 2,402 fail it, although full divisor support exists there at d = 23 and 31. An adaptive policy that tests only shifts with $(d/p) = -1$ succeeds within six selected shifts at every residual prime below 10^6 and fails at p = 1,387,849, whose first six selected shifts 19, 39, 47, 51, 67, 79 all fail both channels and whose seventh, d = 83, succeeds; below 10^7 some primes need eight, nine or eleven. The bound six was learned from development data; the Charter now forbids such a bound the label CONJECTURED until it has survived a disjoint range an order of magnitude larger.

The Galton question, closed. The author asked whether the distribution of potential solutions, viewed from one extreme through the median to the other, carries information the criteria discard. The exact content is this: for fixed x, the centered exponents of a uniformly random divisor of x^2 are independent and uniform on $[-e_i, e_i]$, so the centered support is Binomial(r, 2/3) for squarefree x with r prime factors, and the numbers of candidates permitted by S1 and S2 are $M_1 = 1 + 2\Sigma e_i$ and $M_2 = 1 + 2\Sigma e_i + 4\Sigma_{i<j} e_i e_j$. The decisive negative was found by GPT and

verified by Claude: x = 494,392 = $2^3 \cdot 29 \cdot 2131$ occurs at p = 1,977,361 (d = 207) and at p = 1,977,529 (d = 39), with the same 63 divisors, the same support counts (1, 10, 28, 24) and the same logarithmic sizes, and 0 witnesses in the first case against 5 in the second. No marginal distribution of divisor sizes or supports decides existence; the modulus does. The line is CLOSED as a line of attack; its exact counts are retained as diagnostics.

An empirical per-shift profile. Claude computed, for every residual prime below 10^7 and every index $j \leq 120$, whether a witness of support ≤ 1, ≤ 2 or unrestricted exists at that shift (not only the first), and the author replicated the run in Colab (100.5 CPU-s; identical on every column). Two facts result. First, the unconditional probability that a residual prime has a two-prime witness at index j is 0.33–0.67 for $j \leq 40$ and about 12/j for j = 60–120 (0.211, 0.096, 0.115, 0.128 at j = 60, 80, 100, 120), set by the arithmetic of d — d = 11, 23 and 83 are good shifts — and by nothing resembling a smooth law; multiplied across a window of 576 shifts, the chance that a prime fails everywhere is of order $(12/576)^{12} \approx 10^{-20}$, which is why S2 holds at 20,513 primes with room to spare and why no feasible search can refute it. Second, for one-prime witnesses the hazard — the success rate among primes that failed every earlier shift — falls to about half the unconditional rate beyond $j \approx 15$ (0.096 against 0.152 at j = 15; 0.040 against 0.097 at j = 30), and the only primes still failing at j = 120 are the two genuine S1 failures, 196,561 and 1,977,361; for two-prime witnesses no such memory is measurable, the surviving pool being empty by $j \approx 19$. Shift failures cluster when the witness class is narrow and look independent when it is wide. Two statements of Claude's were corrected by GPT in this exchange and are withdrawn: that the first support-1 witness at 1,977,361 lay "five times" beyond the window (it lies 35% beyond, by index), and that a heuristic failure probability of the form B^{−c log log p} gives a convergent expected number of exceptions for a window B polynomial in log p (it does not: the probability is p^{−o(1)} and its sum over primes diverges). The two facts are consistent: S2 is heuristically false in the limit and unfalsifiable in practice.

The failure-packet model. The one proposal of the round that changes the quantifier asks what the world must look like if every shift in a window fails, and whether the arithmetic of the consecutive integers $x_j = k + j$ can support such a world: a sound relaxation of the genuine all-failure packets (layers: the local failure predicate, exact small-prime valuations, the gcd rule, the global character labels, additional congruences, and realization as an actual factorization), a mutation protocol, and the disposition of every surviving abstract object as realized, excluded or OPEN. The specification PZ-FP-1.0 is frozen and reviewed; the pilot has not been run. A quick look by Claude on 31 October, replicated by the author, calibrates what it will find: on the real packets of the two S1 failures at R = 40, single-prime substitutions that preserve the character label keep the local failure predicate in 93% of cases, the character layer and the gcd layer each reject about three fifths of those, and the congruence that the product equal x_j modulo d_j rejects 99%. The model's power sits in the exact product, not in the cross-shift structure; a simultaneous-obstruction theorem would need an invariant sharper than the gcd and character layers, and none has been proposed. [OPEN. Specification GPT; review and calibration Claude; replication the author.]

5. Are the findings significant, and why

The question the author set for this paper is whether anything above matters, and the answer has to be given in the same labelled vocabulary as the results. We separate four senses of significance.

5.1 Significance for the conjecture: none, and the reasons are explicit

No result here moves the Erdős–Straus conjecture closer to proof, and the line itself proved why its own methods cannot: Theorem U(a′) and Proposition U4 together show that at any fixed shift both failures and successes are infinite in every congruence class, so no finite menu of shifts can be certified, shift by congruence conditions; C3-01 proves that, on the two-free-prime domain, no criterion in p mod d alone decides success, which bounds what a pure residue test can do (whether criteria that retain the individual prime residues extend to three or more non-residue primes is OPEN, Table 4); and §4.6 reduces the residual domain exactly to a pointwise sieve statement that no known analytic method delivers. The 97.60% certification rate is an explanation of known successes, not a measure of progress; the census to $6 \cdot 10^7$ is finite evidence for a bounded statement ($D(p) \leq 107$ below $6 \cdot 10^7$), not evidence for the conjecture. The author's stated working hypothesis — that the systems perform on X and Y but the Z component is absent — was not refuted by this line. Readers who want a proof should stop here. The per-shift profile of §4.11 puts a number on the covering question: a residual prime that has failed every earlier shift still finds a two-prime witness at index j with probability about 12/j, so a bounded window fails with probability about 10^{-20} per prime — which is exactly why a bounded-window universal statement can be neither refuted by search nor proved by the sieve, and why "most primes" and "every prime" remain different claims.

5.2 Mathematical significance: narrow, exact, possibly new, uncertified as original

Six items have a claim to modest mathematical interest. First, A5′ and C3-01 give exact, bounded-size, p-independent residue criteria for success at a shift in the one- and two-free-prime cases, with a precise description of where criteria in p mod d alone end on the two-free-prime domain. The mechanism is classical Jacobi-parity forcing; the statements, their size bounds and the impossibility result for two free primes were not found in the sources consulted. Second, the integral matching theorem is a clean structural statement about the solution set at every prime $p \equiv 1 \pmod 4$ — the bridge map, in every orientation, matches positive solution triples at most one-to-one with no fixed points — with no existence premise, and it is machine-checked. Third, the energy route now has an exact local theory and an exact boundary: Theorem E3 replaces an optimization subproblem by a closed integer formula, Theorem E8 exhausts the inputs on which avoidance is impossible, and Theorems Q1–Q2 show that the universal statement is false and explain where it fails — in the collision-free regime E = M = T — so that the route is known to be a sufficient condition covering all primes outside R and 98.3% of those within it below 10^6, and nothing more. That three exact methods (residue criteria, sieve reduction, energy forcing) all stop on inputs whose shifted numbers have sparse, well-spread divisors is an observation that may deserve a number theorist's second look, with the qualification of §4.8: they are related representations of one divisor condition, and they do not fail on a common set of primes. Fourth, Theorem U(a′), if the cited 2026 theorem on primitively represented primes stands, gives a uniform lower bound for failures at every shift $d \equiv 3 \pmod 4$, extending the shift-7 argument of Dahan to all shifts and to the six classes. Fifth, the exact weight asymptotic of §4.10 answers a

question posed in a 2026 preprint, by an elementary argument, and is the one result of the line whose value does not depend on the conjecture at all; its originality is unchecked and its downstream effect unaudited. Sixth, the simple-witness formulation of §4.11 — Proposition A with a support bound — is a cleaner object than the unrestricted divisor criterion: S2 is cheap to test, exact in its return map, and now carries a measured per-shift profile; its refuted sibling S1 and the shared-x example delimit what the formulation can and cannot see. The channel-depth census, which version 1 counted among the results, is removed from this list by the literature check: the statistic and its record primes are Ionascu and Wilson's. None of these would be publishable in a research journal as a stand-alone advance, and the author does not intend to submit them as such; they are the honest content of three days' directed work, and their value is that each is stated exactly enough to be used or discarded.

5.3 Methodological significance: the record is the result

The stronger claim is methodological. The protocol produced, from two language model systems that are known to generate fluent and confident falsehoods, a record in which every claim carries its hypotheses, its label, its reviewer and its dependencies; in which sixteen stated results were refuted, withdrawn or relabeled with witnesses kept (Table 4: fourteen refuted or withdrawn, two relabeled); in which a tautology dressed as an observation was caught; in which results that a withdrawn system reported as theorems were held as OPEN or unreviewed until a proof arrived and was checked, and only then admitted with the label the check earned; in which a one-hour literature check reclassified a computation the line had treated as its own as a replication of a 2011 paper; and in which a conjecture frozen as a specification at 07:54 on 29 October was refuted by 10:41 on 30 October with two independent implementations, a Proposition-E10 certificate and the author's replication — the decisive experiment being deterministic and costing minutes once the right question (decide, do not sample) was asked; and in which the strongest statements were certified by a proof assistant and rebuilt on infrastructure controlled by the human director alone. The claim-by-declaration review — checking that prose never outruns the Lean — is a discipline the author has not seen described for AI-produced mathematics; no comparison with existing practice has been made and no originality is claimed for it. It found no overstatement. The combination of sealed statement-first exchange, independent validators written blind, exact arithmetic, preregistered counts and the three-valued Lean label did, for these results, work of the kind peer review does, in hours, for results too small to attract reviewers; that version 5 itself needed five corrections from the counterpart system's review (version 6) is the same discipline operating one level up. That is a result about method, and it is reproducible: the repository, the workflow and the census program are public or attached. The Protean Z round added a further instance in the other direction: two of Claude's own statements were corrected by GPT within a day and are withdrawn in §4.11.

5.4 Significance for the author's research program

For the author's larger project on the Computational Human–Machine Conjecture and the "Maestro Model" of directed, source-bound machine scholarship, the line is a case study with a definite shape: the systems can be made to produce correct, bounded, exactly labelled mathematics at the X and Y levels; they can be made to catch each other's errors when the protocol forces blind reconstruction, and to refute each other's conjectures and then confirm the refutation; and they did not, in this line, supply a Z — the one proposal that would have counted (a pointwise sieve result) is one no one has. The supported negative is narrower than version 5

stated it: under the recorded instructions, resources, representations and verification procedures, this line did not produce a verified argument sufficient to settle the conjecture. The experiment does not isolate whether the stop lies in the mathematics, in the systems or in the protocol; a different representation, instruction sequence or allocation of computation could in principle produce a different outcome, and whether it would is an empirical and mathematical question the line leaves open. That the finding can be stated at exactly this width, and no wider, is itself the kind of result the program was designed to produce. Whether a different Z exists on the residual domain remains, as the charter says, an ambition and not a premise.

6. Conclusion and next steps

Between 27 and 29 October 2026 a human-directed line run by two language model systems produced, on the Erdős–Straus conjecture: exact residue criteria that certify 97.60% of an archive of known successes (accounting reconciled from row-level archives in version 6) and, on the two-free-prime domain, provably cannot be extended by conditions on p mod d alone; a set of Lean-certified structural theorems about the solution graph and the fourth-iterate descent, rebuilt independently on the author's account; a conditional density theorem for failures at every shift; an exact local theory of the energy route (closed-form avoidance minimum, collision criterion, complete classification of infinite thresholds, scoped finite-menu theorem) whose universal conjecture was then refuted at 25 primes below 10^6 and, in extended form, at p = 944,329, with the failure located in the collision-free regime; a replication of the Ionascu–Wilson channel-depth census below $6 \cdot 10^7$; an exact reduction of the open classes to a pointwise sieve covering statement that is the known frontier of the analytic method; and, from the review of version 5, an exact asymptotic for a published divisor weight (§4.10), proved and verified but not yet blind-reconstructed or checked for priority. Sixteen claims were refuted, withdrawn or relabeled with witnesses (Table 4), including, from the Protean Z round of 30–31 October, Hypothesis S1 at p = 1,977,361; its sibling S2 holds at every residual prime below 10^7 and is, on the measured per-shift profile, beyond the reach of both search and sieve. Nothing was proved about the conjecture itself, and the paper has tried to say so in every section. The jointly validated artefacts are public.

Next steps, in the order the charter lists them: (1) the Q(p) census, its range segments and the specification erratum are retired as moot; (2) formalization of Lemmas 1–3 of the covering reduction and of the exact minimum formula E3, which are short and carry no analytic dependency; (3) the dependency audit of the 2026 theorem on which U(a′) depends, which can begin from its freely available arXiv text, with the six-class refinement audited separately (§4.4); (4) completion of the literature task — the channel-depth question is now answered (Ionascu–Wilson 2011); the originality of A5′, C3-01 and the energy-route theorems against Bello-Hernández–Benito–Fernández (2026), Xu (2026), Bradford (2021, 2025) and Chamberland (2026) remains to be checked; (5) the p = 1201 discrepancy, now resolved as a misprint in the arXiv version of Ionascu–Wilson (§4.6), to be checked once more against the journal text; (6) the originality search for Theorem W of §4.10 by the citation tool, after which it keeps POSSIBLE_NEW or is relabeled KNOWN, followed by the sealed 128-point pilot and the audit of where the constant enters the cited paper; (7) the method-by-input table of §4.8 over the 1,210 uncertified rows and the 25 failing primes; (8) three research directions proposed in GPT's review, each under a separately frozen XYZ card with its own stop rule — a signed-exponent divisor criterion retaining the individual prime residues for three or more non-residue factors, arithmetic constraints on the avoiding profiles of the energy route (a new predicate, Q itself

remaining REFUTED), and joint constraints among the shifted numbers (p + d_i)/4 of one prime, aimed at the covering question of §4.4; and (9) the director's decision whether the residual-class line continues at all — the one question left standing by §4.2, §4.6 and §4.8 alike being what certifies a firing channel when x_d has few, well-spread divisors — or whether the reduction of §4.6 and the refutation of §4.8 are recorded as the line's terminal results. Added in version 6.3: (10) the PZ-FP-1.0 failure-packet pilot, calibrated in §4.11 and still to be run as specified, with the sharper cross-shift invariant it would need stated as the research question; (11) the development-range rule, adopted 31 October, under which S2's window is a candidate bound until it survives a disjoint range an order of magnitude larger; (12) the director's Colab run of GPT's replication notebook for the S1 refutation and the S2 record.

Appendix A. Proof sketches

A.1 The channel lemmas (classical)

Throughout p is prime, $p \equiv 1 \pmod 4$, $p/4 < x < 3p/4$, $d = 4x - p$; so $d \equiv 3 \pmod 4$, $3 \le d < 2p$ and $\gcd(x, d) = 1$.

Lemma 1 (Type II channel). *Positive integers y, z, both divisible by p, with $4/p = 1/x + 1/y + 1/z$ exist iff some divisor a of x^2 satisfies $a \equiv -x \pmod d$.*

Proof. Write $y = pY$, $z = pZ$. The equation becomes $1/Y + 1/Z = d/x$, i.e. $(dY - x)(dZ - x) = x^2$. Both factors are positive, so $a = dY - x$ is a divisor of x^2 with $a \equiv -x \pmod d$. Conversely, given such a, put $b = x^2/a$; from $ab = x^2$ and $a \equiv -x$ we get $-xb \equiv x^2$, so $b \equiv -x \pmod d$ because $\gcd(x, d) = 1$. Then $Y = (a + x)/d$ and $Z = (b + x)/d$ are positive integers and the identity holds. ■

Lemma 2 (Type I channel). *Positive integers y, z with $p \nmid y$, $p \mid z$ and $4/p = 1/x + 1/y + 1/z$ exist iff some divisor a of x^2 satisfies $a \equiv -px \pmod d$; equivalently, iff some divisor a' of x^2 satisfies $4a' \equiv -1 \pmod d$.*

Proof. With $z = pZ$, clearing denominators gives $(dy - px)(dZ - x) = px^2$. Put $a = dy - px$, $b = dZ - x$; both are positive, $ab = px^2$, $a \equiv -px$ and $b \equiv -x \pmod d$. If $p \mid a$ then $y = p(a/p + x)/d$ is divisible by p, contrary to hypothesis, so $a \mid x^2$. Conversely, given $a \mid x^2$ with $a \equiv -px$, put $b = px^2/a$; then $b \equiv -x \pmod d$ since $\gcd(px, d) = 1$, and $y = (a + px)/d$, $Z = (b + x)/d$ are positive integers with $p \nmid y$. For the second form, $p \equiv 4x \pmod d$ gives $-px \equiv -4x^2$; setting $a' = x^2/a$, which is again a divisor of x^2, $a \equiv -4x^2$ holds iff $a' \equiv -4^{-1}$, i.e. $4a' \equiv -1 \pmod d$. ■

Lemma 3 (the d = 3 channel on the residual domain). *For $p \equiv 1 \pmod{24}$ and $x = (p + 3)/4$, both channels fire at $d = 3$ iff x has a prime factor $q \equiv 2 \pmod 3$.*

Proof. Here $x \equiv 1 \pmod 3$, so $-x \equiv 2$ and $-px \equiv 2 \pmod 3$: both targets are the residue 2. A divisor of x^2 is $\equiv 2 \pmod 3$ iff it contains an odd power of some prime $\equiv 2 \pmod 3$, which exists iff such a prime divides x. ■

Lemma 4 (one-prime sufficient condition). *If x has a prime factor r with $r^j \equiv -x \pmod d$ for some $1 \le j \le 2 \cdot v_r(x)$, Channel II fires at d; with $r^j \equiv -px$, Channel I fires. (Take $a = r^j$ in Lemmas 1–2.)*

A.2 Proof of Theorem A5'

Proof. Every divisor of $x_d^2 = m^2 s^2$ has the form $a \cdot s\text{^}j$ with $a \mid m^2$ and $j \in \{0, 1, 2\}$. Since every prime of m has Jacobi symbol +1, $(a/d) = +1$ for all $a \mid m^2$; and $(s/d) = -1$. Because $d \equiv 3 \pmod 4$, $(-1/d) = -1$, so the Channel I target -4^{-1} has symbol −1 and is hit only by divisors with $j = 1$: $a \cdot s \equiv -4^{-1} \pmod d$. Using $p \equiv 4x_d = 4ms \pmod d$, this reads $a \cdot p \cdot (4m)^{-1} \equiv -4^{-1}$, i.e. $p \equiv -m \cdot a^{-1}$. The set $\{a \bmod d : a \mid m^2\}$ is closed under $a \mapsto m^2/a$, so $\{a^{-1}\} = m^{-2} \cdot \{a\}$ and the condition becomes $p \in -m^{-1} \cdot \{a \bmod d : a \mid m^2\} = -R$. The Channel II target $-x_d = -ms$ has symbol $(-1)(+1)(-1) = +1$, so it is hit only by $j \in \{0, 2\}$. For $j = 0$: $a \equiv -ms \Leftrightarrow 4a \equiv -4ms \equiv -p \Leftrightarrow p \in -4 \cdot \{a\} = -4m \cdot R$. For $j = 2$: $a \cdot s^2 \equiv -ms \Leftrightarrow a \cdot s \equiv -m \Leftrightarrow$ (multiplying by 4m) $p \cdot a \equiv -4m^2 \Leftrightarrow p \equiv -4 \cdot (m^2/a)$, again $p \in -4mR$ since $m^2/a \mid m^2$. Both directions are equivalences, so the criterion is exact, and $S(m, d) = -R \cup -4mR$ has at most $2\tau(m^2)$ elements. The argument uses only that $\gcd(m\ s, d) = 1$ and the Jacobi hypotheses, so d may be composite. ■

A.3 Proof of Lemma U1

Proof. If every prime factor of x_d has Jacobi symbol +1 modulo d, then so does every divisor a of x_d^2, by multiplicativity. The Channel I target -4^{-1} has symbol $(-1/d) = -1$ since $d \equiv 3 \pmod 4$; the Channel II target $-x_d$ has symbol $(-1/d)(x_d/d) = -1$. Neither target can be a divisor residue, so neither channel fires. No hypothesis on p beyond $p \equiv 1 \pmod 4$ is used, and d may be composite. ■

A.4 Proof of Theorem E3 (exact avoidance minimum), condensed from GPT's draft

Proof. An avoiding profile assigns to each surviving complement pair a value (k, k) of mass 2k and energy $2k^2$, to each surviving non-central fixed bin an even value 2k of energy $4k^2$, and to the central bin, if it survives, an odd value $2k + 1$ of energy $(2k + 1)^2$, the baseline 1 being compulsory. Raising one orbit by one unit of two in mass costs $4k + 2$, $8k + 4$ or $8k + 8$ respectively, each sequence increasing in k. The $U = (T - c)/2$ cheapest increments, taken greedily, form initial segments of these sequences, hence a feasible allocation, and any feasible allocation uses U increments of cost at least these, so the greedy total is the minimum. Sorting all increments by cost gives, for each $j \geq 0$, b copies of $8j + 2$, f of $8j + 4$, b of $8j + 6$ and c of $8j + 8$ — a period of $D = 2b + f + c$ increments totalling $8jD + 4D + 4c$. Summing $j = 0, \ldots, q - 1$ gives $4Dq^2 + 4cq$; the remaining $r < D$ increments contribute $(8q + 2)r_1 + (8q + 4)r_2 + (8q + 6)r_3$, and a partial central increment cannot occur since $c \leq 1$ and $r < D$. Adding the baseline c gives the formula. If the central bin was removed, or $D = 0$ with $T > 0$, or $T - c$ is odd or negative, no avoiding profile exists. ■

A.5 Proofs of Theorems E6(b)–(c) and E8, condensed

Proof. E6(b). $x = (p + 7)/4$ is even ($p \equiv 1 \bmod 8$) and $(x/p) = (7/p) = (p/7) = -1$ by reciprocity; $4x \equiv p \pmod 7$ gives $(x/7) = -1$, so $H \ni 2$, x is the full unit group mod 7 and $K = \{1, 2, 4\}$. The central cell is $\{x, 2x, 4x\}$ with ι swapping 2x and 4x ($2 \cdot 4 \equiv 1$); write its occupancies (α, k, k) with α odd and $k \geq 1$ ($x/2$ and $2x$ are divisors of x^2). The target $t_I = -px \equiv -4x \cdot x$, whose ratio to x is $-4x \equiv -p$: central if $p \equiv 6$, in the pair if $p \equiv 3$ or 5. In the latter case removing the pair leaves the central bin alone, so $M = (\alpha + 2k)^2$ and $M - E = 4\alpha k + 2k^2 \geq 6$. E6(c). $x = (p + 15)/4$ is even, $(x/p) = (3/p)(5/p) = (p/3)(p/5) = -1$, and

$x \equiv 7$ or 13 (mod 15), outside $\langle 2 \rangle = \{1, 2, 4, 8\}$; so H is the full unit group, $K = \{1, 4\}$, $x^2 \equiv 4$, $t_I = -4x^2 \equiv 14$, and its cell is $\{11, 14\}$, an ι-pair. A prime $q \mid x$ outside $\langle 2 \rangle$ has q or 2q in the cell and dividing x^2, so the cell has positive mass; avoidance would empty it, so $M = +\infty$. E8. If the target is central, E6(a). Otherwise the central baseline and parities are satisfiable, so any surviving orbit can absorb the remaining even mass, and $M = +\infty$ iff the target orbit (of size ≤ 2) is the whole cell with positive mass, forcing $|K| \leq 2$. Channel I: $-4 \in H$ so $16 \in K$ and $16^2 \equiv 1$ (mod d), $d \mid 255$, $d \in \{3, 15, 51, 255\}$; x even gives $4 \in K$ hence $d \mid 15$, and $p \equiv 1$ (mod 8) gives $d \equiv 7$ (mod 8), so $d = 15$; x odd gives $d \equiv 3$ (mod 8), so $d \in \{3, 51\}$, but $p \equiv 1$ (mod 3) makes $x \equiv 1$ (mod 3), which contradicts $J_d(x) = -1$ at $d = 3$, and at $d = 51$ $x^2 \in K$ gives $x^4 \equiv 1$ (mod 17), so x is a square mod 17 and again $J_51(x) = +1$. At $d = 15$ admissibility forces p mod $5 \in \{2, 3\}$ and E6(c) applies. Channel II: $-x$ is a non-central fixed residue; its orbit is the cell iff $K = \{1\}$; then every generator of H has order ≤ 2, every element of H is a product of distinct prime factors of x, hence a divisor of x^2, and the available target is occupied. ■

A.6 The decision procedure for Q (why it is exhaustive)

Proof. Fix $p \equiv 1$ (mod 24). For each admissible x (non-residue, $(p + 3)/4 \leq x \leq 3(p - 1)/4$) form $d = 4x - p$, factor x, and compute the multiset of residues of the divisors of x^2 from the exponents. If neither target residue $-x$ nor $-px$ occurs, both target bins are empty; the actual histogram restricted to any eligible cell is then itself an avoiding profile of energy E_C (its pair symmetry and parities hold by Lemma E1-type arguments on the involution), so $M_C \leq E_C$ and no cell at this x can force. Otherwise compute H, K, the eligible cosets, T_C, E_C and M_C by Theorem E3 for each; the x forces iff some cell has $E_C < M_C$. Q(p) is true iff some admissible x forces; the scan over all admissible x is finite and complete. Non-firing x are the overwhelming majority (e.g. 16,631 of 16,658 at p = 66,361), which is why the decision costs seconds. The same procedure with the residuosity test removed decides Q-extended. ■

A.7 Proof of Theorem W (exact weight asymptotic), condensed from GPT's derivation

Proof. For coprime u, a write $ua = 2^k \cdot o$ with o odd; then $\varphi(4ua) = 2^{k+1}\varphi(o) = 2ua \cdot \prod_\{q \mid ua, q \text{ odd}\}(1 - 1/q)$, so $1/\varphi(4ua) = h(u, a)/(2ua)$ with $h(u, a) = \prod_\{q \mid ua, q \text{ odd}\}\ q/(q - 1)$, and $W(J) = \tfrac{1}{2} \Sigma\ h(u, a)/(ua)$ over coprime pairs $u, a \leq J$. Write $h(u, a) \cdot [\gcd(u, a) = 1] = \Sigma_\{d \mid u, e \mid a\}\ b(d, e)$. The coefficients b are multiplicative in the pair: multiplying the local generating series of h by $(1 - X)(1 - Y)$ gives, at an odd prime q, the local polynomial $1 + (X + Y)/(q - 1) - (q + 1)XY/(q - 1)$, and at $q = 2$ the polynomial $1 - XY$; the factor at 2 differs from the others and omitting it changes the constant. For $0 < \delta < \tfrac{1}{2}$ the series $\Sigma\ |b(d, e)|(de)^{\wedge}\{\delta - 1\}$ converges, since its odd-prime factors differ from 1 by $O(q^{\wedge}\{-2+2\delta\})$; hence $\Sigma\ b(d, e)/(de)$ converges absolutely, also with logarithmic weights. Exchanging the order of summation, $W(J) = \tfrac{1}{2} \Sigma_\{d, e \leq J\}\ (b(d, e)/(de))\ H_\{\lfloor J/d \rfloor\}\ H_\{\lfloor J/e \rfloor\}$ with H_n the harmonic numbers, and $H_\{\lfloor J/d \rfloor\} = \log J - \log d + O(1)$ uniformly for $d \leq J$. Replacing the product of harmonic numbers by $(\log J)^2$ costs $O(\log J)$ by the absolute logarithmic moments, and extending the coefficient sum to all d, e costs $O((\log J)^2\ J^{\wedge}\{-\delta\})$. The leading constant is therefore $\tfrac{1}{2} \Sigma\ b(d, e)/(de) = \tfrac{1}{2} \cdot (1 - \tfrac{1}{4}) \cdot \prod_\{q \text{ odd}\}(1 + q^{-2}) = \tfrac{1}{2} \cdot \tfrac{3}{4} \cdot (15/\pi^2)/(5/4) = 9/2\pi^2$, using $\prod_q\ (1 + q^{-2}) = \zeta(2)/\zeta(4) = 15/\pi^2$. ■

Claude's check (30 October). The local polynomials follow from $(1 - X)(1 - Y)\cdot(1 + (q/(q - 1))(X/(1 - X) + Y/(1 - Y)))$ at odd q and from $(1 - X)(1 - Y)\cdot(1 + X/(1 - X) + Y/(1 - Y))$ at $q = 2$; their values at $X = Y = 1/q$ are $1 + q^{-2}$ and ¾ respectively, which is the Euler-factor route to the same constant. The numerical behaviour reported in §4.10 was computed with a sieve of $\varphi(4n)$ to $n = 9\cdot10^6$ and a direct double sum; the ratios 0.62795, 0.59503 and 0.57114 at $J = 10^2$, $3\cdot10^2$, 10^3 agree with the cited paper's printed values, which fixes the definition of W(J) used here as Dahan's. The proof does not depend on any downstream result of that paper.

Appendix B. Reproduction

Formal certification. Repository whouxe82/es-lean-certification (public), commit a92f656; GitHub Actions run #1, 28 October 2026, green; workflow .github/workflows/verify-es-lean.yml; Lean 4.19.0 with the pinned Mathlib in lake-manifest.json; axiom log logs/axioms-all.log; audit module lean/Audit.lean; source scan scripts/audit_sources.py. The claim-by-declaration review is a Claude Doc dated 29 October 2026.

Census and witnesses. The program compute.py (standard-library Python, exact integer arithmetic, run with python3 -I) sieves smallest prime factors to $(N + 4000)/4$, and for every prime $p \equiv 1 \pmod{24}$ up to N tests the two channels at $d = 3, 7, 11, \ldots$ by forming the set of residues modulo d of all divisors of x_d^2 from the exponents of x_d. At $N = 6\cdot10^7$ it reports 444,820 primes, 181,503 (40.8%) failing at $d = 3$, maximum depth 107 at $p = 8{,}803{,}369$, in 21.5 seconds; the squares of units modulo 840 are returned as {1, 121, 169, 289, 361, 529}; the witnesses of Table 4 are re-checked by direct divisor enumeration; the identity $4/577 = 1/154 + 1/2308 + 1/177716$ is checked; and the first prime members of the family $528n + 505$ are 1033, 2089, 2617, 3673, 4201, 4729, 6841, 7369. The program and its JSON output accompany this draft.

Energy-route checks. The program check_energy.py (standard library; written from the theorem statements without sight of GPT's code) implements the avoidance minimum by brute-force dynamic programming over orbits and compares it with the closed formula on all 2,050 cases $b, f \in \{0,\ldots,4\}$, $c \in \{0,1\}$, $T \in \{0,\ldots,40\}$ (0 mismatches; bounds, collision threshold and the equality condition also 0 violations); builds H, K, cells, histograms and thresholds from scratch for any (p, x); reproduces the 577 cell (T = 8, E = 12, M = 14) and the 433 example; checks E6(a) on every admissible input at $p < 2{,}000$, E6(b)–(c) on all qualifying primes below 20,000, E8 on every eligible cell at every $p < 1{,}500$, E2's case split on all odd primes below 5,000, the family identities for $n < 200$, and E9 on 13 primes. Runtime about two minutes.

Q decision programs and records. q_attack.py (SHA-256 of the 10^6 version 084a32780510ef144688d1a1f0beb055e521ba95d2e3a20cfbeeac90bd62aaba) decides Q and Q-extended for every prime $p \equiv 1 \pmod{24}$ up to N by the procedure of Appendix A.6; the 10^6 record q_attack_results_1M_claude.json (SHA-256 f5744897525db35904d49881a51526da12dae0b89baaabcb5cfedbfec1b2ba21) lists, for each of the 25 failing primes, every eligible cell at every firing x. Calibration: 29-prime pilot record reproduced exactly; GPT's ES-MC-1.0.0 draws (1,979 cells) reproduced with 0 mismatches. GPT's ES-Q-REF-1 confirmation: report SHA-256 02ff9a5984be6d56ab4ce4692782f61da380b6c248f3a2d3bf02e68314673a5d; certificate 4ef0ba8518bba48477287ca511a358964f79a5ddfd19c5884401b3d053f4b163; evidence archive e29159cabaaa43a5ca22c61d1e306021a9480587b612169c05535a4404851ddd. All are in es-xyz-

validated/ of the public repository with a SHA256SUMS manifest. Weight asymptotic and recovery audit (version 6): verify.py computes the exact identities at p = 66,361, 1,201 and 1,009, the factor-pair search for p = 1201 at x = 301, …, 306, the Table 3 arithmetic, and W(J) to J = 3,000 with the residual diagnostic; audit_rows.py checks the nine recovered CSV lists of §4.3 for duplicates, pairwise overlaps, union, complement and the identity x = (p + d)/4 on every row. Both accompany this version with their outputs. Protean Z (version 6.3): s1s2_claude_v2.py (SHA-256 838d93e3…bdfe) decides S1 and S2 for every residual prime up to N by direct enumeration of signed exponent changes and writes the first shift and witness per hypothesis; its 10^7 record s1s2_10M_v2.csv (d7bcc9f9…044a) holds 20,513 rows, and S1_refutation_certificate_p1977361.json lists, for every $j \leq 441$, the factorization of x_j, the number of witnesses and the minimum centered support. shift_success_profile.py computes the unconditional and conditional (hazard) success rates per shift index for support ≤ 1, ≤ 2 and unrestricted; the author's Colab run to j = 120 carries receipt 725e97a4…. fp_quicklook.py builds the real failure packets of 196,561 and 1,977,361 at R = 40 and scores label-preserving single-prime mutations against the PZ-FP-1.0 layers; the author's Colab run carries receipt 905fed46…. GPT's Protean_Z_Wild_Card.py, its validator and its replication notebook (which embeds Claude's checker and CSV with the hashes above) are in the Protean Z research package.

Director's validation runs. On 29 October 2026 (18:18 ET) the author ran the census program in Google Colab and obtained 444,820 primes, 181,503 failing at d = 3, record depth 107 at p = 8,803,369, the six squares modulo 840, the eight family primes and all witness checks identical to the record, in 34.3 s; at 18:55 ET the energy-route checker, identical; on 30 October at 10:05 and 10:41 ET the Q decision to 120,000 (2.7 s) and to 1,000,000 (876.7 s), identical to Claude's records in every printed figure. Under the Charter these are the director-controlled replications, so Tables 5–6 and Figures 2 and 3 carry the human-validation label.

Governing records. The ES XYZ Research Charter (live document; Word and Markdown exports are snapshots), the Q(p) acceptance record (rev5), the sealed H1/H2 archives of each round with SHA-256 sums, GPT's certification package with its Proof Map and Certification Review PDFs, and the author's working log form the exchange trail. All exchange between systems passed through the author as documents.

Appendix C. Lexicon: translation of key terms

The paper uses the working vocabulary of elementary and analytic number theory, of formal verification, and of the project's own protocol. Table 7 translates each term into plain language for readers outside those fields, including mathematicians in other areas. The translations are deliberately informal; the exact definitions are those given where each term first appears (the last column), and nothing in this appendix changes a statement in the body. Terms are listed in roughly the order in which the paper needs them: the conjecture and its reduction, the divisor criteria, the energy route, the formal and analytic dependencies, the evidence vocabulary, and the computational record.

Table 7. Lexicon of key terms.

Term	Plain-language translation	Where
Erdős–Straus conjecture	The claim (1948) that for every whole number $n \geq 2$ the fraction $4/n$ can be written as a sum of three fractions with numerator 1, such as $4/5 = 1/2 + 1/4 + 1/20$. Still unproved.	title, §1
unit fraction; Egyptian fraction	A fraction with numerator 1 ($1/x$). A sum of distinct unit fractions is an "Egyptian fraction", after the ancient Egyptian way of writing fractions.	§1
prime p	A whole number greater than 1 divisible only by 1 and itself. It is enough to prove the conjecture for primes, so the paper works only with them.	throughout
$a \equiv b \pmod m$; residue; residue class	"a and b leave the same remainder when divided by m." The remainder is the residue; all numbers with the same remainder form a residue class. "$p \equiv 1 \pmod 4$" means p is one more than a multiple of 4.	throughout
the six square classes mod 840; residual classes; six-class prime	Divide a prime by 840 and look at the remainder. Classical identities settle the conjecture for every remainder except six (1, 121, 169, 289, 361, 529), which are exactly the perfect squares among the possible remainders. Primes with those remainders are the only ones still open; the paper calls them six-class or residual primes.	§1, Table 1
polynomial identity; the polynomial obstruction	A formula giving the three denominators as fixed polynomial expressions in n, valid for every n in a residue class. A classical theorem says no such formula can exist for the six square classes; that is why they are hard. The obstruction says nothing about rules that choose denominators case by case.	§1, §4.6
shift (general sense)	The paper's own word for the odd number d added to the prime p before dividing by 4. Each shift moves the search to a different "shifted number" $x_d = (p + d)/4$, and the method asks, shift by shift, whether x_d^2 has a divisor of the required kind. Only about log p shifts need examining before a solution is expected, which is why the census, the window B(p) and the hypotheses S1, S2 are all stated in terms of how many shifts a prime must try.	§2, §4.6, §4.11
admissible shift d	An odd number d that leaves remainder 3 on division by 4, with $d < 2p$. Adding it to p and dividing by 4 produces a candidate smallest denominator x. Every solution of $4/p$ corresponds to some d.	§2, Table 1
$x_d = (p + d)/4$	The candidate first denominator produced by the shift d. The whole method studies the divisors of x_d^2.	§2
divisor	A number that divides another exactly (3 is a divisor of 12). The criteria ask whether x_d^2 has a divisor with a prescribed remainder modulo d.	throughout
channel (general sense)	A route from a divisor of x_d^2 to a solution of $4/p = 1/x + 1/y + 1/z$: a divisibility condition that a divisor a must satisfy, together with a return map that turns a into the remaining denominators y and z. Proposition 2.1 shows there are exactly two such routes, distinguished by how many of the three denominators are multiples of p. "Channel" is not a standard term of number theory; the paper uses it to keep the two cases apart in proofs, in the Lean library and in the census, where an incidence is a (p, d, channel) triple.	§2, §4.3, App. A.1
Channel I / Channel II; Type I / Type II solution	The two ways a solution can exist at a given x: Channel II (Type II) needs a divisor a of x_d^2 with $a \equiv -x_d \pmod d$ and gives a solution in which two denominators are multiples of p; Channel I (Type I) needs 4a	§2, App. A.1

Term	Plain-language translation	Where
	≡ −1 (mod d) and gives one with exactly one such denominator. Proposition 2.1 says these are the only two possibilities.	
a channel "fires"; success / failure at d	A channel fires at d when the required divisor exists, which yields an explicit solution. Success at d means some channel fires; failure means neither does.	§2
return map	The explicit formulas that turn a successful divisor a into the actual denominators y and z, so that a certificate can be checked by multiplying out.	§2, App. A.1
channel depth D(p)	The smallest shift d at which a channel fires for the prime p — equivalently, how far one must look before finding a solution of the standard shape. The census measures its distribution.	§4.6
covering statement	The reformulation of the conjecture on the hard classes: for every such prime, some shift of bounded size fires. Proving it for every prime is exactly what is open.	§4.6
quadratic residue / non-residue	r is a quadratic residue modulo d if r is the remainder of some perfect square divided by d; otherwise a non-residue. Whether the prime factors of x_d are residues or non-residues modulo d decides most of the paper's criteria.	§2, §4.2
Legendre symbol; Jacobi symbol (r/d)	A compact notation taking the value +1 (residue) or −1 (non-residue). The Legendre symbol is for prime d; the Jacobi symbol extends it to composite d by multiplying the Legendre symbols of d's prime factors.	Table 1, §4.2
quadratic reciprocity	A classical theorem (Gauss) relating "is p a square modulo q" to "is q a square modulo p". Used to move information between the moduli p and d; imported from the Lean library as an audited dependency.	§3.3, §4.2
QR-smooth	Every prime factor of x_d is a quadratic residue modulo d. Lemma U1: such x_d can never succeed at d.	Table 1, §4.2
one-free-prime; two-free-prime (A5′, C3-01)	Cases in which x_d has exactly one, or exactly two, prime factors that are non-residues modulo d. In these cases the paper gives exact tests (Theorems A5′ and C3-01) for whether a channel fires.	§4.2
the sets R, S(m, d), W	Finite lists of remainders modulo d, computed from the residue part m of x_d, such that the test "does a channel fire" reduces to "is a certain remainder in the list". Their sizes are bounded by the number of divisors of m^2.	§4.2
τ(n), ω(n), φ(n)	τ(n): the number of divisors of n. ω(n): the number of distinct prime factors of n. φ(n) (Euler's totient): how many numbers from 1 to n share no factor with n.	Table 1, §4.10
chain saturation; exponent box; subgroup H	A group-theoretic test: the remainders of the prime factors of x_d generate a set H of remainders that is closed under multiplication (a subgroup). If the exponents in x_d are large enough ("saturate" the chain), every element of H is a divisor remainder, so a channel fires exactly when its target lies in H.	§4.2
coset; cell; subgroup K = H^2	In the energy route the remainders in H are sorted into "cells" (cosets of the squares K of H): two remainders are in the same cell if their ratio is a square in H. Targets and divisors are compared cell by cell.	§4.7
energy (general sense)	A sum of squares of counts. If w(r) is the number of divisors of x_d^2 landing in remainder bin r, the energy of a cell is Σ $w(r)^2$. The name is	§4.7

Term	Plain-language translation	Where
	borrowed from additive combinatorics, where “additive energy” counts coincidences among sums, and ultimately from physics, where energy grows with the square of an amplitude. Here it measures how unevenly the divisors are spread over the bins: an even spread has the least possible energy, and piling divisors into few bins raises it. Energy is a whole number computed exactly from the factorization of x_d; no estimate is involved.	
divisor histogram w(r); mass T; energy E	Count how many divisors of x_d^2 fall into each remainder bin r; that is the histogram w. Its mass T is the total count in a cell; its energy E is the sum of the squares of the counts (large when divisors pile up in few bins).	§4.7
involution; complement pair; central residue; fixed bin	The pairing $a \leftrightarrow x^2/a$ of divisors maps remainder bins onto each other. A bin paired with a different bin forms a complement pair; a bin paired with itself is a fixed bin; the bin of x itself is the central residue.	§4.7
target; avoiding profile; avoidance minimum M; forcing	The target is the bin a divisor must occupy for a channel to fire. An avoiding profile is any hypothetical histogram with the same mass and symmetries that leaves the target empty; M is the least energy such a profile can have. If the real energy E is below M, no avoiding profile can match the real one, so the target must be occupied: the solution is "forced".	§4.7
finite energy; finite-energy forcing theorem	“Finite” records that the whole energy argument is a finite integer computation: a cell has finitely many bins, the counts are whole numbers, and the avoidance minimum M is the least energy over a finite list of candidate profiles, so M is a whole number (or $+\infty$ if no avoiding profile exists), never a limit or an estimate. The finite-energy forcing theorem (Lean declaration energy_lt_avoidanceMinimum_forces) is the abstract statement that any histogram with the required symmetries whose energy is strictly below M must place a divisor in the target bin. Lean proves it for arbitrary finite index sets; the step identifying the divisors of x_d^2 with such a histogram is a separate obligation, checked arithmetically in each instance (for example p = 577).	§3.4, §4.7, Table 2
collision	Two distinct divisors landing in the same bin. The collision count determines the energy; the "collision-free regime" (every bin holds at most one divisor) is where the energy test can never force anything.	§4.7, §4.8
Conjecture Q; Q-extended; Q′	Q: at every open prime some admissible x has a cell where forcing happens. Q-extended: the same without the non-residue restriction on x. Q′: the residual form after the proved reductions. All three are refuted (false) in §4.8, while the conjecture itself holds at every failing prime.	§4.7–4.8
certificate (Proposition E10)	A complete, checkable record: for a positive claim, the actual denominators; for a refutation of Q at a prime, an explicit avoiding profile for every eligible cell, so that anyone can re-verify the verdict without trusting the search program.	§4.7, §4.9
witness; divisor witness; first witness	A concrete object that demonstrates a claim by inspection, as opposed to an argument that such an object exists. For a positive claim the witness is the divisor a of x_d^2 (equivalently the coprime pair u, v with $a/x_d = u/v$) from which the return map rebuilds the denominators; the first witness for a prime is the one at the smallest shift, and the centered support of a witness counts the primes whose exponents differ between a and x_d. For a refutation the witness is the counterexample itself — a prime, a	§3.3, §4.5, §4.11, Table 4

Term	Plain-language translation	Where
	shift, a histogram — that makes the claim false; REFUTED verdicts keep their witness in Table 4. An H_N witness is a prime at which the hypothesis H_N holds. Every numerical witness in the paper was re-verified by direct arithmetic.	
bridge map T; B1, B4; integral matching theorem	A rational transformation that carries one solution of 4/p to another. B1 and B4 are its first and fourth iterates. The matching theorem says that, restricted to whole-number solutions, this map pairs solutions at most one-to-one and never with themselves.	§4.1
descent classification; normalized Type II source (δ, r, s, t)	A standard way of writing a Type II solution with four parameters, and a theorem listing exactly which such solutions can return to whole numbers under B4.	§4.1
H_D, H_N	Two unproved hypotheses, each of which would imply the conjecture if true. They are stated as definitions in the formal library and never asserted.	§4.1
family p = 528n + 505; Obláth's criterion	An arithmetic progression of numbers every prime member of which has a solution of a known shape; and the classical observation that if p + 1 has a prime factor leaving remainder 3 on division by 4, a solution exists immediately.	§4.1, §4.7
Dirichlet's theorem	Every arithmetic progression a, a + m, a + 2m, … with gcd(a, m) = 1 contains infinitely many primes. Used whenever the paper says "infinitely many primes in this class"; an audited dependency.	§3.3, §4.2
binary quadratic form; primitively represented	An expression such as $u^2 + uv + kv^2$. A number is represented by the form if it equals the form at some whole numbers u, v, and primitively represented if those u, v share no factor. The 2026 theorem (FHRSS) counts primes represented this way.	§4.4
sieve; sieve methods; exceptional set	Techniques that count the numbers surviving a list of divisibility conditions, descended from the sieve of Eratosthenes for listing primes. The paper uses the word in two roles. As a method (Vaughan 1970; Pomerance and Weingartner 2026) it shows that primes without a solution of the standard shape are rare: the “exceptional set” has density zero. As a target (§4.6) it names what a proof would need: a pointwise sieve statement that among the roughly log p shifted numbers (p + 3)/4, (p + 7)/4, … at least one has a prime factor in a prescribed class, for every prime p rather than for almost all. Known sieve methods lose a positive proportion of the primes at each step, so they deliver the almost-all version but not the pointwise one; that gap is the paper's stated reason the six classes remain open. (A third, purely computational use: the census program “sieves” smallest prime factors as a bookkeeping step.)	§4.4, §4.6
density; ≫ N/(log N)^{3/2}; O(·); o(1)	Ways of saying how a count grows. "≫ N/(log N)^{3/2}" means at least a constant times that quantity for large N; O(f) means bounded by a constant times f; o(1) means a quantity tending to zero. "Density zero" means a vanishing proportion, which is compatible with infinitely many members.	§4.4, §4.6, §4.10
conditional; CONDITIONAL on X	True provided the named published theorem X is true. The line uses the label whenever it relied on a published proof it did not re-verify.	§3.3
audited dependency	A published theorem the author ruled may be used without the CONDITIONAL label because its proof is formalized in the Lean library (Dirichlet, reciprocity).	§3.3

Term	Plain-language translation	Where
PROVED / FINITE_VERIFIED / CONJECTURED / REFUTED / INCONCLUSIVE	The evidence labels. PROVED: a complete written proof reviewed by the other system. FINITE_VERIFIED: checked by exact computation up to a stated bound only. CONJECTURED: proposed, unproved. REFUTED: false, with the counterexample kept. INCONCLUSIVE: no verdict.	Table 1, §3.3
KNOWN / EQUIVALENT_TO_KNOWN / POSSIBLE_NEW / NOT_ASSESSED	The novelty labels: already in the literature; a restatement of something known; not found in the sources consulted (which is not a proof of novelty); not checked.	Table 1
Lean 4; Mathlib; STATED / COMPILED / VERIFIED	Lean is a computer proof assistant that checks every step of a proof mechanically; Mathlib is its standard mathematical library. STATED: the theorem was written in Lean but not built. COMPILED: built and checked on one system's machine. VERIFIED: rebuilt from the pinned sources on the author's own GitHub account.	§3.4, §4.1
axiom closure; whitelist; sorry	A Lean proof may silently rest on extra assumptions (axioms). The audit lists every axiom each theorem uses and rejects anything beyond the three standard ones; "sorry" is Lean's placeholder for an omitted proof and is forbidden.	§3.4
claim-by-declaration review	A check, sentence by sentence, that the prose of the certification package claims no more than what the Lean declarations prove.	§3.4, §4.1
sealed; SHA-256 hash	A file is sealed by recording its SHA-256 hash — a 64-character fingerprint that changes if a single byte changes — so that later readers can confirm the file is the one that was used. A hash proves identity of bytes, not truth.	§3.2
statement-only exchange; blind reconstruction	The receiving system is given a theorem's statement without its proof and must produce its own proof or a recorded failure before seeing the original. Agreement reached this way is stronger evidence than agreement after reading.	§3.2, §4.10
producer; validator; independent implementation	The producer is the program that searches for or computes a result; the validator is a separately written program that checks the result without using the producer's code. "Independent" is always qualified by what was kept separate.	§3.4, §4.9
director; director-controlled rerun	The author's role: he decides what is launched and labelled, and reruns the frozen programs himself on infrastructure he controls as the final check.	§3, App. B
XYZ card; X, Y, Z	The protocol's way of posing a task: X, what is already established; Y, the exact obstacle to going further; Z, the proposed change of representation, question, constraint or scale that addresses Y.	§3.1
census; (p, d) row; incidence; unit of count	A census is an exhaustive computation over every prime below a bound. A (p, d) row is one prime–shift pair; an incidence counts a (p, d, channel) triple; the paper always names which unit a count uses because the same data give different totals under different units.	§3.3, §4.3, §4.6
W4 archive; B6 residual	The project's development data: all six-class primes below ten million against fifteen prime shifts (W4), and the 50,515 successful rows within it that an earlier test had failed to explain (B6), on which the residue criteria were measured.	Table 1, §4.3

Term	Plain-language translation	Where
Monte Carlo	Estimating by random sampling rather than exhaustive enumeration. Useful for finding patterns or counterexamples; never a substitute for a complete check.	§4.9
Ionascu–Wilson classes C_i	A published (2011) classification of primes by how far one must search for the first denominator — the same quantity as the channel depth, which is why the census is a replication.	§4.6
W(J); weight; Lemma 4.22	A weight is a number attached to each term of a sum so that some terms count for more than others; a weighted sum replaces "how many" by "how much". The divisor weight W(J) of Dahan's 2026 preprint (his Lemma 4.22) sums $1/\varphi(4ua)$ over coprime pairs u, a ≤ J: each pair contributes the reciprocal of a totient, so pairs with few prime factors weigh more. In his argument the weight controls how densely a family of search witnesses covers the primes, and its leading constant enters his depth–density trade-off. Theorem W of §4.10 gives the exact growth, $W(J) = (9/2\pi^2)(\log J)^2 + O(\log J)$, with $9/2\pi^2 \approx 0.456$, three times the constant of the cited lower bound.	§4.10
asymptotic; $(\log J)^2$; Euler product; multiplicative convolution	An asymptotic describes how a quantity behaves for very large inputs. W(J) grows like a constant times $(\log J)^2$. The constant is computed by factoring the sum prime by prime (an Euler product), after rewriting it as a sum over divisor pairs (a multiplicative convolution).	§4.10, App. A.7
centered support; u/v; S1, S2	Write a divisor witness as a ratio u/v relative to x itself; the centered support counts how many distinct primes had their exponents changed. S1 and S2 are the claims that a witness of support at most 1 or 2 always exists within a short window of shifts.	§4.11
window B(p); index j	The number of candidate shifts a hypothesis is allowed to try, here $\min((p-1)/2, b^2)$ with b the number of binary digits of p (441 for a prime around two million); the index j counts shifts, $d = 4j - 1$ is the shift itself.	§4.11
per-shift success rate; hazard; shift memory	The fraction of primes with a witness at a given shift (unconditional), and the same fraction among primes that failed every earlier shift (hazard). Hazard below the unconditional rate means failures cluster — a prime that has failed many shifts is more likely to fail the next.	§4.11
failure packet; abstraction; mutation	The set of consecutive shifted integers $x_j = k + j$ for one prime, with their factorizations, viewed as a single object; the model relaxes it to abstract "atoms" and asks whether a packet in which every shift fails could exist at all. A mutation changes one atom to test which constraint rejects the result.	§4.11
Kneser's theorem; quotient budget	A classical lower bound for the size of a product of subsets of an abelian group. Proposition B turns it into a checklist: if the prime factors of x contribute enough in every quotient of the group they generate, all residues are reached and a channel fires.	§4.11

Table 7 is a reading aid, not a glossary of record: where a plain-language rendering and a definition in the body differ, the body governs.

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Bibliographic data were checked by Claude against arXiv, publisher and Erdős Problems pages on 29 October 2026, and independently re-checked by a citation tool (Perplexity) on 30 October 2026, which confirmed 32 of 34 entries and supplied ten corrections or additions adopted above. Two of its findings were examined further and not adopted as stated: the Mathlib Dirichlet declaration does exist (see that entry); the second author of the 10^{18} verification is given with full names pending confirmation. Entry 26 remains an unverified antecedent. A second citation-tool check on 31 October 2026 (version 6.3) confirmed the pre-2026 entries and the Kneser and DeVos entries, and asked for the alphabetical placement of the Gao and Kneser entries (done) and for a live check of the eleven 2026 arXiv and journal entries before posting; its one substantive proposal, to change the Elsholtz–Tao page range to 501–523, was not adopted — the published range is 50–105.